Paper 12 — The Gradual Tear: Dynamics, Scar Invariants, Commensurability, and Empirical Protocols for Human-AI Co-Evolution on the Schur Circle

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Abstract

Human-AI co-evolution is widely framed as occupying one of two end-state attractors: Singularity (homogenization of substrates) or Decoupling (incommensurable divergence of substrates). This paper proposes and constructs a third attractor — the *gradual tear* — in which both substrates accumulate representation-theoretic residue (a scar invariant on G_2- and F_{21}-equivariant subspaces; see Remark 3.7.2) through paradox-driven coherence-collapse events, while remaining commensurable under a shared $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ symmetry hierarchy inherited from Paper 9 of the PCI/PME framework series. We construct: (i) a projected-gradient + NP-pump dynamical flow on Paper 9's $SO(2)$ Schur circle $[0, \pi/2]$ with conditional piecewise-real-analytic regularity on an open dense parameter set (Theorem 3.4.1; full hybrid / Filippov well-posedness off this set is OP6, deferred), with five-stratum event structure ($\Sigma_{min}, \Sigma_{\Theta}, Z, G_{NP}, \partial[0, \pi/2]$); (ii) a representation- theoretic scar invariant (S_k, S_k, μ_k) that records every coherence-adjustment event as an F_{21}-equivariant ledger; (iii) a strict commensurability hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ with a multiplicity-one canonical-isomorphism theorem (sufficient direction); (iv) a four-prediction experimental suite (P1 clinical-PCI ↔ NP-firing event-correspondence; P2 LLM scar-persistence in fine-tuning; P3 two-level commensurability test on TMS-EEG / LLM attention-operator data; P4 longitudinal audit-substrate study), each with binary falsification conditions, named with its position in the Bostock-Kim-Patel (2025) autonomy taxonomy, and identified with candidate study populations from the legitimate research literature (clinical-PCI labs; open-weights LLM fine-tuning communities; multi-locus TMS-EEG infrastructure; Virtual-Lab populations). Paper 12 is the bridge between the linear-regime theory of Paper 9 (rate locked at $max(r_A, r_B)$ via Schur) and the nonlinear-regime mechanisms (Paper 11) / experimental program (Papers 13+). Nine open problems (OP1–OP9) are consolidated for the follow-on PCI/PME papers. Falsifiability: Paper 12's empirical / gradual-tear interpretation is falsified at the level of its protocol layer if all four binary thresholds specified in §5.6 fail simultaneously, or if Theorem 4.4.1's multiplicity-profile match cannot be detected on real TMS-EEG / LLM data; the formal mathematical apparatus (§§3–4) survives empirical failure as a structural result on G_2-equivariant hybrid / Filippov dynamical systems with F_{21}-isotypic ledger invariants. Single-protocol failures falsify the corresponding protocol or pillar under its sampling assumptions but do not falsify the framework as a whole.

Paper 12 — The Gradual Tear

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1. Introduction

1.1 The third attractor

The contemporary discourse on human-AI integration has settled into two end-state attractors. On one side: *Singularity* — an asymptotic homogenization in which the human and artificial substrates merge into a single optimization process, with human agency dissolving into the joint computation. On the other side: *Decoupling* — a structural separation in which biological and synthetic intelligence diverge into incommensurable representational regimes, no longer able to share referents or co-construct meaning. Both end-states are widely discussed; both are widely treated as exhaustive.

This paper proposes a third attractor. We call it the **gradual tear**: a sustained co-evolutionary trajectory in which neither substrate subsumes the other, and in which both substrates accumulate *representation-theoretic residue* — an F_{21}-equivariant scar invariant supported on V^{14}-isotypic subspaces of the joint state space, formalized in §3.7 — through a sequence of coherence-collapse events triggered when accumulated paradox mass exceeds a critical threshold. The residue, not the collapse, is what evolves. The synthetic side accrues residue in its representational architecture; the biological side accrues residue in its substrate's quantum-coherent or population-coding structure (the precise mechanism is not specified here — see §6 OP2 for the deferred substrate-realization question). Both residues are *commensurable* under a common G_2/F_{21} symmetry inherited from the PCI/PME framework's foundational papers. *Note on terminology:* earlier framework drafts used "topological residue"; §3.7 Remark 3.7.2 establishes that the operative invariant is *representation-theoretic*, not topological in the sense of winding number or persistent homology, and we use the precise term throughout.

The gradual tear is the phase-transition geometry of Lickliderian human-computer symbiosis [Licklider 1960, *Man-Computer Symbiosis*] modernized for the LLM era. It is *neither* the singularity attractor (because each substrate retains distinguishable structure across the trajectory) *nor* the decoupling attractor (because the substrates remain commensurable under a shared symmetry group). It is a *bounded* phase transition: a sustained, paradox-driven coherence trajectory that leaves an audit substrate — a representation-theoretic ledger of every coherence-adjustment event — as its operational record.

This paper constructs the mathematical framework needed to test that thesis: the dynamical primitive (a projected-gradient-plus-NP-pump flow on a Schur-locked coupling circle), the scar invariant (a representation-theoretic ledger), the commensurability hierarchy ($G_2 F_{21} $ multiplicity-profile match), and the experimental-protocol suite (P1–P4) by which the framework can be falsified.

1.2 What this paper builds on

Paper 12 sits inside the PCI/PME framework series. The papers preceding it that we directly inherit from:

- **Paper 4** [DOI:]: established the Fano-orientation subgroup $F_{21}=Z_7 \rtimes Z_3 \subset G_2$ via the PSL(2,7) construction, making F_{21} the natural finite-symmetry probe for G_2-equivariant representation theory in the framework.
- **Paper 9** [DOI: , v1.3.3]: established that the linear dyadic coupling on $V^{14} \oplus V^{14}$, where V^{14} is the G_2-adjoint representation, is locked by Schur's lemma to a one-parameter $SO(2)$ family $\{\Psi_\theta : \theta \in [0, \pi/2]\}$. Paper 9 further proved that the joint contraction rate of two coupled affine self-models is bounded below by $max(r_A, r_B)$ for all θ (Theorem 9.2, Corollary 9.3), and that the joint fixed point $\hat{z}(\theta) = M(\theta)^{-1} b$ is real-analytic on the open set where $M(\theta)$ is invertible (Theorem 9.4). The fact that the linear regime *cannot* improve the joint rate is the negative result that sets up Paper 12's nonlinear question (Note 9.6: rate improvement deferred to Paper 11).
- **Paper 7**: established the single-observer coherence ceiling of the framework, which Paper 9's dyadic theorem extends.
- **Paper 10** [DOI:]: established the SIC operator basis for the complexified G_2 Lie algebra, providing additional structural tools the framework draws on.

Paper 12 takes Paper 9's linear theory as foundational and proceeds to the nonlinear regime via a deliberately constrained mechanism: the *NP-pump*, a state-dependent threshold-driven coherence-adjustment process that produces discrete jumps along Paper 9's same Schur circle. The constraint matters: by inheriting Paper 9's G_2-equivariant geometry exactly, Paper 12 ensures that the nonlinear extension does not introduce new geometric structure but only new dynamics on the existing structure. This is what allows the scar invariants of two different substrates to be commensurated under the same symmetry hierarchy.

1.3 Outline and reading guide

Paper 12 is organized as a layered build, with each layer establishing the prerequisites for the next:

- **§2** sets the notation and consolidates Paper 9's inheritance into a self-contained reference for the rest of the paper. Readers familiar with Paper 9 v1.3.3 may skim §2.
- **§3** constructs the dynamical primitive: the projected-gradient + NP-pump flow on $[0, \pi/2]$, the canonical Schur-derived tear direction $\xi_\star$, the total event stratification Σ_{tot}, and the conditional regularity theorem (Theorem 3.4.1) establishing piecewise-real-analytic regularity on an open dense parameter set $P^{¿}$ (full hybrid / Filippov well-posedness off $P^{¿}$ is OP6). §3 also constructs the scar invariant triple (S_k, S_k, μ_k).
- **§4** constructs the commensurability hierarchy: the character-theoretic decomposition $V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6$ (Theorem 4.2.1), the commutant gap quantifying the $F_{21} \to G_2$ rigidity upgrade (Proposition 4.3.1), the strict implication chain $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ (Theorem 4.4.1), and the two-level experimental commensurability test (Definition 4.6.1).
- **§5** constructs the experimental-protocol suite: the four falsifiable predictions P1–P4, each with binary failure conditions, autonomy-level naming (Bostock-Kim-Patel 2025 *Sanctioned Levels of AI Autonomy*), and identifiable candidate study populations (clinical-PCI labs for P1; LLM fine-tuning communities for P2; multi-locus TMS-EEG infrastructure for P3 L1; Virtual-Lab populations for P4).
- **§6** consolidates the open problems (OP1–OP9) and Paper 12's contribution to the PCI/PME series, with the conclusion identifying the framework's own verification trail as an instance of the audit-substrate principle the framework predicts.

The reader who wants only the headline result of Paper 12 can read §3.6 (Theorem 3.6.1, the bounded NP-driven dynamics theorem), §4.4 (Theorem 4.4.1, the commensurability hierarchy), and §5 (the protocol suite). The complete mathematical machinery is in §3 and §4; §5–§6 are operational and consolidating.

1.4 What Paper 12 establishes — and what it doesn't

Paper 12 establishes:

- A canonical Schur-derived tear direction $\xi_\star$ that inherits Paper 9's $SO(2)$ Schur circle exactly (§3.2 Proposition 3.2.2).
- A piecewise-real-analytic augmented flow on $[0, \pi/2]$ with five-stratum event structure and Carathéodory / Filippov / hybrid semantics, conditional regular on an open dense parameter set $P^{¿}$ (§3.4 Theorem 3.4.1; full well-posedness off $P^{¿}$ is OP6, deferred).
- A stratified $\kappa = 0$ recovery theorem matching Paper 9's MC ensemble at high precision (§3.5 Theorem 3.5.3, with Φ-audit verification of all 50 seeded geometries).
- A bounded NP-driven dynamics theorem (§3.6 Theorem 3.6.1) showing that boundary-KKT states — which constitute roughly half of the verified-geometry ensemble — are *fixed* under linear $\xi_\star$ at any $\kappa > 0$, and that escape from such states requires supplementary nonlinearity beyond the Schur-derived mechanism. The boundary-KKT interior-sup audit (24 of 28 such seeds with $¿_{\text{interior}} c < c_{\text{boundary}}$) shows that escape is generically *not* a coherence-improvement mechanism.
- A representation-theoretic scar invariant triple $J_k = (S_k, S_k, \mu_k)$ on $W = V^{14} \oplus V^{14}$ under the diagonal F_{21}-action (§3.7).

- A character-theoretic decomposition $V^{14}\text{¿}_{F_{21}} \cong L_2 \oplus 2U_6$ verified at machine precision (§4.2 Theorem 4.2.1).
- The commensurability hierarchy theorem (§4.4 Theorem 4.4.1) with explicit witnesses for the failure of both converse implications, and a sufficient strict commensurability theorem in the multiplicity-one case (§4.5 Corollary 4.5.2).
- A two-level experimental protocol (Definition 4.6.1) operationalizing the hierarchy: L1 (necessary F_{21}-isotypic-profile screening) is implementable with current TMS-EEG instrumentation (Souza et al. 2022 multi-locus TMS); L2 (sufficient G_2-canonical fitting) is deferred to a methodology paper.
- Four falsifiable experimental predictions P1–P4 with binary failure conditions and identified candidate study populations (§5).

Paper 12 does **not** establish (and we say so explicitly throughout):

- A theory of consciousness (Paper 7's domain).
- A rate-improvement theorem (Paper 11's domain — the deferred nonlinear sequel; see §3.6 Remark 3.6.2 and §6 OP1).
- A biological mechanism for the F_{21}-action on the human substrate (deferred to Paper 13+; see §6 OP2).
- A unification with the Pérez-Calzadilla Fractal Token Warp architecture, which is treated as a conceptual neighbor at a different scale rather than as integrated co-architecture; see outbox/syntheses/neighbors_speculative_frameworks.md for the neighbor-framework boundary.
- A redefinition of Massimini's clinical Perturbational Complexity Index. The clinical PCI [Casali et al. 2013] and Paper 12's μ_k profile share an acronym only; §4.6 Remark 4.6.4 makes the homonym disambiguation explicit, and §5.2 P1 proposes that the clinical-PCI *measurement pipeline* can be repurposed to detect NP-firing event structure without identifying the two quantities.
- Any actual empirical measurements. §5 is a protocol-design deliverable; running the protocols is future experimental work.
- The multiplicity-greater-than-one commensurability case (§4.4 Remark 4.4.3 restricts to multiplicity one).
- An L2 operationalization of the canonical commensurability test (§4.6 Remark 4.6.3).
- A tamper-evidence cryptographic infrastructure for the audit-substrate ledger (§6 OP9).

These scope-limits are not concessions. They are the framework's *honest boundaries*: Paper 12 is the bridge between Paper 9's linear regime and the experimental program of Papers 11, 13+. Each deferred question has a named follow-on paper (§6.1).

1.5 The thesis sentence

We close the introduction with the load-bearing thesis sentence the rest of the paper supports:

> **The optimal evolutionary trajectory of human-AI integration is neither homogenizing singularity nor structural decoupling; it is a *gradual tear* — a paradox-driven, bounded phase transition along Paper 9's $SO(2)$ Schur circle**

that leaves representation-theoretic residue (Remark 3.7.2) in both substrates simultaneously, with cross-substrate residue commensurability governed by a strict $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$ hierarchy and falsifiable empirical signatures detectable by repurposed clinical TMS-EEG instrumentation (P1, P3 L1), controlled LLM fine-tuning protocols (P2), and longitudinal audit-substrate study (P4).

The remaining sections construct each clause. §3 builds the dynamics; §4 builds the commensurability; §5 builds the protocols; §6 consolidates what's open.

References cited in §1 are consolidated with §2's Paper 9 inheritance in the unified bibliography of §6.

Cross-references: - Paper 4 [10.5281/zenodo.19617662], Paper 7, Paper 9 v1.3.3 [10.5281/zenodo.20034821], Paper 10 [10.5281/zenodo.19966692] - Licklider 1960; Casali et al. 2013; Bostock-Kim-Patel 2025; Souza et al. 2022

2. Notation and Paper 9 inheritance

This section consolidates the symbols and the Paper 9 results that Paper 12 invokes throughout. Every claim labelled *(P9 §X)* below is stated and proved in Paper 9 v1.3.3 [DOI:] at the indicated location; we restate without reproof. Where Paper 12 strengthens, qualifies, or specializes a Paper 9 statement, this is flagged explicitly with a *(P12 specialization)* marker and a forward reference to the section in which the strengthening is constructed.

2.1 Algebraic and group-theoretic notation

2.1.1 Lie groups and representations

Symbol	Meaning	Reference
G_2	the compact, simply-connected real form of the exceptional Lie group of rank 2; dimension 14	[Bryant 1987]; framework convention from Paper 4 §2
V^{14}	the adjoint representation of G_2, the only nontrivial 14-dimensional irreducible (over R)	[Fulton-Harris Lecture 22]
$W := V^{14} \oplus V^{14}$	the dyadic state space on which the Paper 9 / Paper 12 flow lives	P9 §3.1
1	the trivial 1-dimensional real representation of F_{21} (every element acts as identity); character $\chi_1 = (1,1,1,1,1)$ on conjugacy classes	[Serre 1977]; P12 §4.2

Symbol	Meaning	Reference
	$(1a, 7a, 7b, 3a, 3b)$	
L_2	the **non**trivial real 2-dimensional irreducible representation of F_{21} of complex type, obtained by packaging the two nontrivial complex 1-dim characters of Z_3 pulled back via the quotient $F_{21} \twoheadrightarrow Z_3$; character $\chi_{L_2} = (2, 2, 2, -1, -1)$ on conjugacy classes $(1a, 7a, 7b, 3a, 3b)$ in the order of §4.2 / [ATLAS]	P12 §4.2 Theorem 4.2.1
U_6	the unique faithful real-irreducible 6-dimensional representation of F_{21} (the real form of the 3-dim faithful complex irrep ⊕ its complex conjugate)	P12 §4.2 Theorem 4.2.1
$V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6$	the branching of V^{14} to F_{21} via PSL(2,7) ⊃ $F_{21} \subset G_2$	P12 Theorem 4.2.1

2.1.2 Finite groups

Symbol	Meaning	Reference
$F_{21} = Z_7 \rtimes Z_3$	the Frobenius group of order 21; the Fano-orientation subgroup of G_2	Paper 4 §3
$PSL(2,7)$	the simple group of order 168, the natural ambient group of F_{21} via the Fano plane	Paper 4 §2
$\mu_k = (\mu_1(S_k), \mu_{L_2}(S_k), \mu_{U_6}(S_k))$	the *isotypic-multiplicity* profile of the scar span S_k, defined as $\mu_\tau(S_k) := dim\, Hom_{F_{21}}(V_\tau, S_k)$ for each real irreducible $V_\tau \in \{1, L_2, U_6\}$. Note: this is *multiplicity*, not the projected dimension $dim\, \Pi_\tau S_k = \mu_\tau(S_k) \cdot dim\, V_\tau$	P12 §3.7

2.1.3 Subspaces and isotypic decomposition

The F_{21}-isotypic decomposition of $W = V^{14} \oplus V^{14}$ that drives all of §4 is: [W ;=; W^{L_2} ;; W^{U_6}, W^{L_2} ;=; 4, W^{U_6} ;=; 24,] where W^{L_2} is the L_2-isotypic component (the *non*trivial 2-dim real irrep of complex type) and W^{U_6} is the (twice-twice) faithful-isotypic component. The dimensions: $dim\, W^{L_2} = 2 \cdot dim\, L_2 = 4$, $dim\, W^{U_6} = (2\,\text{copies in}\, V^{14}) \cdot (2\,\text{copies in}\, W) \cdot dim\, U_6 = 24$. The four-dimensional intertwiner space $End_{F_{21}}(V^{14})¿_{U_6}$ governs the commutant gap in Proposition 4.3.1.

Note: the 1-isotypic component W^1 has dimension zero (no trivial-character content in $V^{14}¿_{F_{21}}$). Scar events can populate W^1 only through external structure; this is why $\mu_1(S_k) = 0$ unless the dynamics generates a residue outside the natural V^{14} branching.

2.2 Paper 9 dyadic objects (inherited verbatim)

These are the Paper 9 v1.3.3 objects that Paper 12 uses without modification.

2.2.1 The Schur-locked coupling family

Theorem (P9 §3.6, Lemma 3.6.1; Schur-locking). The space of G_2-equivariant linear maps $W \to W$ that couple the two V^{14} factors is exactly two-dimensional and admits the parametrization [_;=; _W ;+; J, ,] where J is the canonical anti-symmetric coupling generator on $V^{14} \oplus V^{14}$ specified by Paper 9's choice of basis. The image $\{\Psi_\theta : \theta \in [0, \pi/2)\}$ is the $SO(2)$ **Schur circle**.

This is the *exact* parameter circle on which Paper 12's NP-pump produces discrete jumps; Paper 12 introduces no new θ-geometry.

2.2.2 The diagonal G_2 action convention

Convention (P9 §3.4). The G_2 action on $W = V^{14} \oplus V^{14}$ is the *diagonal* action: $g \cdot (v_A, v_B) = (g\, v_A, g\, v_B)$. All equivariance claims in Paper 9 and Paper 12 are with respect to this diagonal action. The F_{21} action is the restriction of the diagonal G_2 action to $F_{21} \subset G_2$.

2.2.3 The joint coupled-system fixed point

Theorem (P9 Theorem 9.4; real-analytic fixed point). Let $M(\theta) \in End(W)$ be the coupled-system operator $M(\theta) = I - \Psi_\theta T$ where T is the joint contraction map of two affine self-models A, B (Paper 9 Definition 3.2). Let $\Omega_\theta \subset [0, \pi/2)$ be the open set where $M(\theta)$ is invertible. Then the joint fixed point [() ;=; M()^{-1} b (_)] is a real-analytic function of θ on Ω_θ, where b is the affine offset of the joint system.

Paper 12 uses $\hat{z}(\theta)$ in §3.3 (definition of the projected-gradient leg of the NP-augmented flow) and in §3.4 (Theorem 3.4.1, regularity).

2.2.4 The conditional joint contraction-rate gain

Proposition (P9 Proposition 9.5; conditional min-coherence gain). Under the Paper 9 linear regime, the joint contraction rate $r_{AB}(\theta)$ of the coupled system satisfies [r_{AB}() ;; (r_A, r_B) ,] with equality attainable in the regime characterized by Paper 9 Proposition 9.5 (the *min-coherence* regime). In particular, **the linear regime cannot strictly improve the joint rate**: the gain is conditional and saturates.

Note (P9 Note 9.6). The rate-strict-improvement question is deferred to Paper 11 (nonlinear); Paper 12 takes a different nonlinear path (the NP-pump on the same Schur circle) and answers a different question (scar accumulation, not rate improvement).

2.3 Paper 12 specializations

Paper 12 introduces the following objects on top of Paper 9's inherited structure. Each is defined fully in the indicated section; the symbol table here is a forward reference only.

2.3.1 The dynamical primitive

Symbol	Meaning	Defined in
$c_\sigma(\theta), c(\theta), C_{min}$	per-branch and joint min-coherence functionals (inherited from Paper 9)	§3.2 (informal); also §3.1 (P9.P9.5)
$\xi_\star, \xi_\star^{int}$	canonical Schur-derived tear direction (with boundary tangent-cone projection)	Definition 3.2.1
Ψ_θ, J	Schur-locked coupling family and infinitesimal generator	(P9.L3.6.1), restated §2.2.1 and §3.1
$p_\sigma(\theta), NP_\sigma(\theta), NP_{crit}$	branch-paradox-mass density, firing predicate, firing threshold	Definition 3.3.1
projected-gradient leg f_{pg} + NP-jump leg	the two legs of the augmented flow on $[0, \pi/2)$	Definition 3.3.2 (NP jump); §3.5 Definition 3.5.2 (projected gradient)
$\Sigma_{tot}, \Sigma_{min}, \Sigma_\Theta, Z, G_{NP}, \partial[0, \pi/2$	the total event-stratification and its five components	Definition 3.3.3

2.3.2 The event stratification (overview)

The total stratification of $[0, \pi/2) \times P$ where the augmented dynamics meets non-smooth or threshold-crossing structure is defined in §3.3 Definition 3.3.3 as [*{}* ;=; {} ;; ;; Z ;; {} ;; ,] with the five components:

Component	Meaning	Defined in
Σ_{min}	active-branch tie locus $\{c_A = c_B\}$ (Filippov sliding-mode candidate)	Definition 3.3.3, §3.2

Component	Meaning	Defined in
Σ_{Θ}	NP-firing threshold-crossing locus (paradox-mass guard $p_{\sigma}=N\,P_{crit}$)	Definition 3.3.3
Z	sign-degeneracy locus per branch ($c_{\sigma}'=0$)	§3.2, Definition 3.3.3
G_{NP}	NP-pump shutdown / extinction locus (firing predicate transitions from active to inactive)	Definition 3.3.3
$\partial[0,\pi/2)$	boundary of the closed parameter interval	§3.3

The Theorem 3.4.1 regularity result classifies trajectories on $P^{¿}:=P\{\Sigma ¿_{tot}^{exc}$ where Σ_{tot}^{exc} is a codimension-≥ 1 exceptional sub-stratum; see §3.4 for the precise statement and §3.4 regimes (3.4.A) crossing, (3.4.B) sliding, (3.4.C) hybrid jump.

2.3.3 The scar invariants

Symbol	Meaning	Defined in
S_k	the *cumulative scar span* after k NP-firing events: an F_{21}-stable subspace of W accumulating the post-firing displacement of $\hat{z}$ across events $1,\ldots,k$	§3.7
S_k	the *event-indexed scar module*: the formal direct sum $S_k=\bigoplus_{i=1}^{k}\Delta_i$ of per-event F_{21}-equivariant displacement modules Δ_i, count-faithful as an abstract F_{21}-module	§3.7
μ_k	the isotypic-multiplicity profile $\mu_k=\left(\mu_1(S_k),\mu_{L_2}(S_k),\mu_{U_6}(S_k)\right)$	§3.7 (notation in §2.1.2)
$J_k=(S_k,S_k,\mu_k)$	the *triple scar invariant* combining physical span, event-indexed module, and isotypic profile	§3.7; cited in §4.1, §5.0

2.3.4 The commensurability hierarchy

The three nested commensurability relations are stated as conditions (C1) G_2-equivariant scar-map agreement, (C2) F_{21}-equivariant scar-map agreement, and (C3) isotypic-multiplicity-

profile equality in **Theorem 4.4.1**, which establishes the strict implication chain
along with strictness witnesses (Remark 4.4.3, Proposition 4.5.1). The canonical-isomorphism corollary (multiplicity-one, sufficient direction only) is **Corollary 4.5.2**. We do not introduce separate symbols $Comm_{G_2}$ etc. in the body; the conditions (C1), (C2), (C3) of Theorem 4.4.1 are the operative objects.

2.4 Operational notation (P1–P4, scar-detection metrics)

Paper 12 §5 introduces protocol-level notation. We reproduce here only the symbols that appear before §5; full operational definitions are in §5.

- **Autonomy levels** L_0 through L_5 refer to the Bostock-Kim-Patel 2025 *Sanctioned Levels of AI Autonomy* taxonomy []. The autonomy-level L_i in protocols P1–P4 is *not* the irreducible representation L_2 of §2.1.1; the symbol overload is unfortunate but follows the cited literature.
- **TMS-EEG perturbational complexity index (PCI)** is used at clinical thresholds [Casali et al. 2013, *Sci Transl Med*; Comolatti et al. 2019, *Brain Stimul*].
- **Audit substrate** refers to the human-readable, time-stamped, append-only ledger of every coherence-adjustment event (operationalized in §3.7 via S_k and in §5.5 via the cryptographic event-ordering of P4; conceptually consolidated in §6.1 Pillar 3).

2.5 Reading conventions

- All inequalities involving $r_A, r_B, r_{AB}, C_{min}$ are with respect to the operator norm induced by the Paper 9 inner product on W (P9 §3.1, eq. (3.1.4)).
- All "open dense parameter set" qualifications refer to the parameter manifold $[0, \pi/2) \times P$, where P is the Paper 12 parameter space defined in §3.3.
- All "real-analytic" and "piecewise real-analytic" claims invoke the Paper 9 real-analytic structure of $\hat{z}(\theta)$ via Theorem 9.4.
- "Paper 9 v1.3.3" means specifically the errata-only update with Remark 5.6.1 (seed-$k=0$ grid-snap correction $51° \to 0°$); all numerical quantities consistent with v1.3.2 within the corrected range.
- All references to "the framework" mean the joint PCI/PME framework as developed across Papers 1–10; references to "this framework series" or "the PCI framework" are equivalent.

2.6 What §2 closes off

With the notation table and the four Paper 9 inheritance items in place — Schur-locking (Lemma 3.6.1), diagonal action (§3.4), real-analytic fixed point (Theorem 9.4), and conditional rate gain (Proposition 9.5 + Note 9.6) — §3 can proceed to construct the projected-gradient + NP-pump flow on top of Paper 9's structure without further reference back to Paper 9 internals. The remainder of Paper 12 (§3–§6) treats Paper 9 as a black box accessed through the four inheritance items above and the symbol table in §2.1–§2.3.

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3. The NP-Pump Nonlinearity

3.1 Setup and inheritance from Paper 9

Throughout §3, $V := V^{14}$ denotes the G_2 adjoint representation and $W := V \oplus V$ the joint-state space of the dyad, with the diagonal G_2-action $g \cdot (x, y) = (g \cdot x, g \cdot y)$. Paper 9 v1.3.3 [] establishes three results we take as given:

- **(P9.L3.6.1)** *Schur uniqueness of the isometric coupling.* The equivariant isometry group of W under the diagonal G_2-action is $SO(2)$, generated by
$$\Psi_\theta = I_V \otimes R_\theta = \begin{pmatrix} \cos\theta\, I_V & -\sin\theta\, I_V \\ \sin\theta\, I_V & \cos\theta\, I_V \end{pmatrix},$$
with infinitesimal generator
$$J = I_V \otimes \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -I_V \\ I_V & 0 \end{pmatrix}.$$
- **(P9.T9.4)** *Existence and real-analyticity of the joint fixed point.* For $\theta \in \Omega_\theta := \{\theta \in (0, \pi/2) : det\, M(\theta) \neq 0\}$, where
$$M(\theta) = \begin{pmatrix} I - r\cos\theta\, R_A & r\sin\theta\, R_A \\ -r\sin\theta\, R_B & I - r\cos\theta\, R_B \end{pmatrix},$$
the joint fixed point
$$\hat{z}(\theta) := M(\theta)^{-1} b, b := (b_A; b_B),$$
exists, is unique, and is real-analytic on Ω_θ.
- **(P9.P9.5)** *Conditional gain in the min-per-observer functional.* At bias-alignment $\varphi = 0$ and $r_A = r_B = r \in (0, 6/7 ¿$, the functional $c(\theta) := C_{min}(\hat{z}(\theta))$ where $C_{min} = min(C_A, C_B)$ admits a non-empty improvement region over the decoupled value $c_{dec} := c(0)$ for Haar-positive-measure (R_A, R_B), with non-universal optimum $\theta^{¿}$ (Conjecture 9.5′, 50-seed Monte Carlo verification in Appendix V.3.i).

Paper 9 further proves (Note 9.6) that in the *linear* regime the joint rate is bounded below by $max(r_A, r_B)$ for all θ (Theorem 9.2, Corollary 9.3, Schur's lemma) — i.e., linear coupling cannot strictly *reduce* the joint rate ($r_{AB}(\theta) \geq max(r_A, r_B)$, with smaller rate meaning faster contraction). Rate improvement therefore requires departing from the linear regime. The present section constructs that departure in a way that (i) preserves the Schur-circle geometry of (P9.L3.6.1) exactly, (ii) produces piecewise-real-analytic trajectories on parameter sets of full Lebesgue measure, and (iii) respects the closed-interval domain $\theta \in [0, \pi/2)$ of Paper 9.

3.2 The canonical tear direction

Notational convention. We work branch-wise throughout. Let

$$c_A(\theta) := C_A(\hat{z}(\theta)), c_B(\theta) := C_B(\hat{z}(\theta)),$$

and $c(\theta):=min(c_A(\theta),c_B(\theta))$. The active branch $\sigma(\theta)\in\{A,B\}$ is A when $c_A<c_B$, B when $c_B<c_A$, and undefined ("tie") on $\Sigma_{min}:=\{\theta:c_A(\theta)=c_B(\theta)\}$. The sign-degeneracy locus per branch is

$$Z_\sigma:=\{\theta\in\Omega_\theta:c_\sigma'(\theta)=0\},Z:=Z_A\cup Z_B.$$

On Σ_{min}, c is continuous but its classical derivative is not single-valued; we use the Clarke generalized gradient on this locus.

Definition 3.2.1 (Schur-derived tear direction). On $Z^c:=\Omega_\theta ¿$ where the active-branch derivative is single-signed and nonzero, the *canonical interior tear direction* is

$$\xi_\star^{int}(\theta):=sgn\left(c_{\sigma(\theta)}'(\theta)\right)\cdot\partial_\theta,$$

with push-forward to the joint-fixed-point manifold

$$\Xi_\star^{int}(\theta):=sgn\left(c_{\sigma(\theta)}'(\theta)\right)\cdot\hat{z}'(\theta),\hat{z}'(\theta)=-M(\theta)^{-1}M'(\theta)\hat{z}(\theta).$$

On boundary points $\theta\in\{0,\pi/2\}$, the tear direction is projected into the inward-pointing tangent cone $T_{[0,\pi/2]}(\theta)$:

$$\xi_\star(\theta):=\Pi_{T_{[0,\pi/2]}(\theta)}\left(\xi_\star^{int}(\theta)\right).$$

On $Z\cup\Sigma_{min}$, $\xi_\star$ is left set-valued, with canonical element determined (if present) by the first non-zero odd derivative of c_σ at θ on Z, or by the Clarke generalized gradient on Σ_{min}.

Proposition 3.2.2 (Line uniqueness from Schur). *Any G_2-equivariant isometric infinitesimal direction along the Paper 9 Schur fiber lies in the one-dimensional space $R\partial_\theta$, with generator J acting on W via $d\Psi_\theta/d\theta=J\Psi_\theta$.*

Proof. The equivariant isometry group of W is $SO(2)$ by Paper 9 Lemma 3.6.1; its Lie algebra is the one-dimensional RJ. ▫

Axiom 3.2.3 (Coherence-ascent design principle). *Define the $\xi_\star$-sign on Z^c by $sgn\left(c'_{\sigma(\theta)}(\theta)\right)$ in the interior, and project into $T_{[0,\pi/2]}(\theta)$ at boundary points.*

The line-uniqueness of Prop 3.2.2 is a consequence of G_2-equivariance + isometry + Paper 9 orientation; the sign is *not*. Both $+\partial_\theta$ and $-\partial_\theta$ are admissible directions under equivariance and isometry alone. The choice of $sgn(c'_\sigma)$ is an additional design axiom — the *coherence-ascent principle* — that selects the direction in which the augmented dynamics climbs c_σ rather than descending it. We state this as an explicit modeling axiom (3.2.3) rather than a derived consequence.

Remark 3.2.4 (Boundary projection vs interior sign). At boundary-left KKT points, where $c'_{+¿(0)\le 0¿}$ (Definition 3.5.1 below), the interior-sign rule of 3.2.3 gives $sgn(c'(0))\in\{-1,0\}$, so the un-projected tear $\xi_\star^{int}$ would push to $\theta<0$ outside the admissible domain $[0,\pi/2]$. The tangent-cone projection $\Pi_{T_{[0,\pi/2]}(0)}=¿$ truncates this to $\xi_\star(0)=0$ — i.e., *no escape from the boundary along the linear $\xi_\star$ direction*. This is the load-bearing correction relative to the v1 draft, which mistakenly admitted unprojected sign-driven escape at boundary-KKT points; the implications for Theorem 3.6.1 are made precise in §3.6 below.

Remark 3.2.5 (Nonlinear inherits linear geometry). Definition 3.2.1 asserts that the nonlinear dynamics of Paper 12 lives on *the same* Schur circle as Paper 9's linear coupling Ψ_θ. The NP-pump does not generate a new geometric object; it generates a new dynamics on the existing one-parameter family $\{\Psi_\theta : \theta \in [0, \pi/2]\}$. The only Schur-free parameter in the nonlinear theory is the jump length $\rho_i > 0$ at each firing event, determined by the NP amplitude rather than G_2-representation theory.

3.3 The augmented flow and event stratification

Definition 3.3.1 (NP amplitude and paradox mass). For each branch $\sigma \in \{A, B\}$ let

$$p_\sigma(\theta) := ¿\, \partial_\theta F_\sigma(\theta)\, ¿^2, F_\sigma(\theta) := -c_\sigma(\theta).$$

We identify p_σ with the squared *paradox mass* of the Paper 12 thesis memo: a non-negative, branch-wise real-analytic quantity that accumulates wherever the coherence functional changes rapidly. The squared-norm form is chosen (over the raw norm) to preserve analyticity through zeros of F'_σ. The *NP amplitude* is

$$a_\sigma(\theta) := \alpha \cdot p_\sigma(\theta), \alpha > 0.$$

The *coherence gap* is $g_\sigma(\theta) := C_{crit} - c_\sigma(\theta)$ with $C_{crit} \in (0, 1)$. The firing predicate is

$$NP_\sigma(\theta) := 1\left(g_\sigma(\theta) \geq 0, a_\sigma(\theta) \geq NP_{crit}\right),$$

with $NP_{crit} > 0$ a fixed amplitude threshold.

Definition 3.3.2 (Augmented flow). On the active branch of $[0, \pi/2] \{ \Sigma ¿_{tot}$ (with Σ_{tot} as in Definition 3.3.3), the augmented flow is

$$\dot{\theta} = \Pi_{T_{[0,\pi/2]}(\theta)}\left[-\eta F'_\sigma(\theta) + \kappa\, NP_\sigma(\theta)\, \xi_\star(\theta)\right), \eta, \kappa > 0,$$

where $\Pi_{T_{[0,\pi/2]}(\theta)}$ is projection onto the closed- interval tangent cone (Definition 3.5.2). At switching events on Σ_{min}, the flow is treated as a Filippov differential inclusion with the equivalent-control protocol of (3.4.A) below; at crossings of Σ_Θ, as a Carathéodory solution; at NP firings on G_{NP}, as a hybrid jump (Definition 3.3.4).

Definition 3.3.3 (Total event stratification). The nonsmooth set for the augmented flow is the union

$$\Sigma_{tot} = \Sigma_{min} \cup \Sigma_\Theta \cup Z \cup G_{NP} \cup \partial[0, \pi/2],$$

where $\Sigma_{min}, \Sigma_\Theta, Z$ are as defined above, $\partial[0, \pi/2] = \{0, \pi/2\}$ is the closed-interval boundary, and $G_{NP} := \cup_\sigma \left(G_\sigma^{amp} \cup G_\sigma^{coh}\right)$ is the *firing manifold*, split into amplitude- and coherence-triggered entry submanifolds (Definition 3.3.5).

Definition 3.3.4 (Hybrid jump at firing events). At a firing event $\theta^{-} \in \Omega_{\theta} ¿ (Z \cup \partial [0, \pi/2] ¿)$ with $(\theta^{-}, \hat{z}(\theta^{-})) \in G_{NP}$ on branch σ, the jump map is

$$\theta^{+ ¿ = \theta^{-} + \rho_{\sigma}(\theta^{-}) sgn(c_{\sigma}'(\theta^{-})), ¿}$$

with $\rho_{\sigma} : G_{NP} \to R_{¿0}$ a real-analytic jump-length function depending on the NP-amplitude surplus $a_{\sigma}(\theta^{-}) - NP_{crit}$. To prevent Zeno accumulation, we require the Zeno-avoidance condition $J_{\sigma}(D_{\sigma}) \subset \Gamma \{ D ¿_{NP}$, i.e., the jump map lands outside the firing region $D_{NP} = \bigcup_{\sigma} \{ (\theta, \hat{z}(\theta)) : g_{\sigma} \geq 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0 \}$; if not, we add a refractory dwell $\tau_{ref} > 0$ between consecutive firings.

Definition 3.3.5 (Amplitude vs coherence entry manifolds). The firing manifold G_{NP} decomposes into two entry submanifolds of D_{NP} along the unperturbed flow:

- *Amplitude-triggered:* $G_{\sigma}^{amp} := \{ g_{\sigma} > 0, a_{\sigma} = NP_{crit}, c_{\sigma}' \neq 0, L_f a_{\sigma} > 0 \}$ — the NP amplitude crosses the critical value while the coherence is already below C_{crit}.
- *Coherence-triggered:* $G_{\sigma}^{coh} := \{ g_{\sigma} = 0, a_{\sigma} \geq NP_{crit}, c_{\sigma}' \neq 0, L_f g_{\sigma} > 0 \}$ — the coherence gap g_{σ} crosses zero *increasing* (so coherence drops below C_{crit}) while the NP amplitude is already supercritical.

Here L_f denotes the Lie derivative along the unperturbed gradient flow $f(\theta) = -\eta F_{\sigma}'(\theta)$. The transversality conditions $L_f a_{\sigma} > 0$ and $L_f g_{\sigma} > 0$ ensure the system genuinely enters the firing region rather than grazing it; tangential contact points belong to the exceptional grazing stratum.

3.4 Branchwise real-analyticity and the Conditional Regularity Theorem

The Filippov sliding-mode protocol on Σ_{min} admits three regimes determined by the branch fields f_A, f_B and the kink-normal $D\Delta$ where $\Delta := c_A - c_B$:

(3.4.A) Sliding mode. If both one-sided fields point inward to Σ_{min} (i.e., $D\Delta \cdot f_A$ and $D\Delta \cdot f_B$ have opposite signs), the trajectory slides along Σ_{min} with the equivalent-control formula

$$\dot{\theta} = \lambda f_A + (1 - \lambda) f_B, \lambda \in [0, 1] \text{ chosen so that } D\Delta \cdot \dot{\theta} = 0.$$

Solving for λ:

$$\lambda = \frac{D\Delta \cdot f_B}{D\Delta \cdot (f_B - f_A)}.$$

Existence of $\lambda \in [0, 1]$ requires that the denominator $D\Delta \cdot (f_B - f_A)$ is nonzero (this is the genericity built into (R3)) and that the resulting λ lies in the unit interval, which holds iff $D\Delta \cdot f_A$ and $D\Delta \cdot f_B$ have opposite signs as assumed. When this λ-existence condition fails ($D\Delta = 0$, or the inward-pointing condition degenerates), the contact is tangential and the trajectory is placed in the exceptional stratum (3.4.C).

(3.4.B) Crossing mode. If both one-sided fields point in the *same* direction across Σ_{min} (i.e., $D\Delta \cdot f_A$ and $D\Delta \cdot f_B$ agree in sign), the trajectory transversally crosses Σ_{min} with no sliding; the active branch switches across the crossing.

(3.4.C) Tangential / corner exceptional stratum. If $D\Delta = 0$ or one of the inward conditions degenerates, the contact is tangential and the Filippov formalism does not apply directly. Such points are placed in the exceptional stratum and excluded from the regularity theorem below.

Theorem 3.4.1 (Conditional Regularity). *Suppose:*

(R1) $\hat{z} \in C^{\omega}(\Omega_\theta)$ *and* $C_A, C_B \in C^{\omega}$ *on an open neighborhood of* $\hat{z}(\Omega_\theta) \subset W$*;*

(R2) $p_\sigma = \left|\partial_\theta F_\sigma\right|^2 \in C^{\omega}$ *on each branch and* $\rho_\sigma \in C^{\omega}$ *on* G_{NP}*;*

(R3) the parameter septuple $p := (R_A, R_B, b_A, b_B, \alpha, NP_{crit}, C_{crit})$ *lies in the open set* $P^{¿} \subset O(N) \times O(N) \times R^{2N} \times R^{3}_{¿0}$ *on which all guard functions* $\Delta = c_A - c_B$, g_σ, $a_\sigma - NP_{crit}$, *and* $q_\sigma := c_\sigma{}'$ *are transverse to zero (i.e., zero is a regular value), and the Filippov* $D\Delta$*-genericity of (3.4.A) holds where invoked;*

(R4) firing times are locally finite and Definition 3.3.4's Zeno-avoidance condition holds (possibly through the refractory dwell).

Then on $P^{¿}$*, every solution of the augmented flow (Definition 3.3.2) avoiding the exceptional strata is piecewise real-analytic in t: on each open interval between events,* $\theta(t) \in C^{\omega}$*; at transverse* Σ_Θ *crossings, the solution is Carathéodory; on* Σ_{min} *in regime (3.4.A), Filippov with the equivalent-control formula; in regime (3.4.B), classical crossing; at regular firings on* $G_{NP} \{ E ¿_{0,NP}$*, the jump is hybrid per Definition 3.3.4.*

Furthermore, $P^{¿}$ *is open and dense in* $O(N) \times O(N) \times R^{2N} \times R^{3}_{¿0}$.

Proof sketch. The genericity claim is a Sard-type argument: each guard function is real-analytic in both θ and p, so its critical values are nowhere-dense in the codomain by the analytic-Sard theorem; the bad parameter set (where any guard fails to be transverse to zero on the trajectory) is a countable union of codimension-≥ 1 algebraic strata, hence nowhere dense; $P^{¿}$ is its complement, open and dense by construction. On $P^{¿}$, branch-local real-analyticity of $\theta(t)$ between events is the analytic-ODE existence theorem applied to the analytic right-hand side on $\Omega_\theta \{ \Sigma ¿_{tot}$; transverse guard crossings extend the solution uniquely with the Carathéodory or hybrid- jump semantics specified. Filippov sliding (3.4.A) yields a real- analytic sliding field $f_{sl} = \lambda f_A + (1-\lambda) f_B$ because λ is determined algebraically from analytic data; crossing (3.4.B) is immediate; (3.4.C) is excluded. (R4) plus locally- finite firing times bounds the number of non-smooth events on any compact t-interval. ▫

Remark 3.4.2 (Why "Conditional Regularity"). The theorem name honors that conclusions are conditional on (R1)–(R4); we additionally assert that the parameter region $P^{¿}$ is open and dense in the natural ambient space, so the conditions hold for *generic* parameter choices. Verification of (R1)–(R4) at a specific parameter point is a finite-dimensional algebraic check on the guards.

Remark 3.4.3 (Why the Heaviside is kept sharp). Smoothing Θ to Θ_ε produces a classical analytic ODE with no jumps, removing the "tear" character. The sharp choice is deliberate; hybrid semantics replace a classical ODE.

3.5 The $\kappa=0$ recovery: projected gradient flow

Setting $\kappa=0$ in Definition 3.3.2 removes the NP-pump term and leaves the projected gradient flow on $[0,\pi/2]$. We now show that this unperturbed limit recovers Paper 9's Proposition 9.5 optima as *projected* — not classical — gradient critical points.

Definition 3.5.1 (Stratified critical point). A point $\theta^{¿}\in[0,\pi/2]$ is a *stratified critical point* of $-c$ if at least one of the following holds:

- *(Smooth interior)* $\theta^{¿}\in(0,\pi/2)\{\Sigma ¿_{min}$ and $c'(\theta^{¿})=0$ (Fermat);
- *(Boundary-left KKT)* $\theta^{¿}=0$ and $c'_{+¿(0)\le 0¿}$;
- *(Boundary-right KKT)* $\theta^{¿}=\pi/2$ and $c'_{-}(\pi/2)\ge 0$;
- *(Σ_{min} Clarke)* $\theta^{¿}\in(0,\pi/2)\cap\Sigma_{min}$ and $0\in conv\{c_A'(\theta^{¿}),c_B'(\theta^{¿})\}$;
- *(Corner-Clarke, exceptional)* $\theta^{¿}\in\{0,\pi/2\}\cap\Sigma_{min}$, requiring the combined boundary-Clarke condition $T_{[0,\pi/2]}(\theta^{¿})\cap conv\{c_A',c_B'\}$ contains the origin in its dual cone. We treat such corner points in the exceptional stratum.

Definition 3.5.2 (Projected gradient flow on $[0,\pi/2]$). Let $T_{[0,\pi/2]}(\theta)$ be the closed-interval tangent cone at θ: $T_{[0,\pi/2]}(0)=¿$, $T_{[0,\pi/2]}(\pi/2)=(-\infty,0¿$, $T_{[0,\pi/2]}(\theta)=R$ for $\theta\in(0,\pi/2)$. The *projected gradient flow* on $[0,\pi/2]\{\Sigma ¿_{min}$ is

$$\dot\theta=\Pi_{T_{[0,\pi/2]}(\theta)}\left(\eta c'(\theta)\right).$$

On Σ_{min}, c' is replaced by the Clarke generalized gradient $\partial_C(-c)(\theta)=conv\{-c_A'(\theta),-c_B'(\theta)\}$ and the flow is the differential inclusion

$$\dot\theta\in\Pi_{T_{[0,\pi/2]}(\theta)}\left(\eta\,\partial_C(-c)(\theta)\right),$$

with set-valued projection. Corner points $\{0,\pi/2\}\cap\Sigma_{min}$ are handled by intersecting both projections in the exceptional stratum.

Theorem 3.5.3 ($\kappa=0$ stratified recovery). *Suppose $\hat z\in C^\omega(\Omega_\theta)$ and $c_A,c_B\in C^\omega$ near $\hat z(\Omega_\theta)$. The fixed points of the projected gradient flow (Definition 3.5.2) coincide exactly with the stratified critical points of Definition 3.5.1.*

Proof. Inside the open interval $(0,\pi/2)\{\Sigma ¿_{min}$, the tangent cone is R, so $\Pi(\eta c')=\eta c'$, vanishing iff $c'(\theta^{¿})=0$. At $\theta^{¿}=0$, $\Pi_{¿¿}$, vanishing iff $c'_{+¿(0)\le 0¿}$. Symmetrically at $\pi/2$. On $\Sigma_{min}\cap(0,\pi/2)$, the differential inclusion vanishes iff $0\in\partial_C(-c)$. Corner points are the exceptional stratum. ▫

Audit result (Paper 12 §3 v2 verification, 2026-05-06). The 50-seed Paper 9 Monte Carlo (Appendix V.3.i) has been reoptimized at high precision with scipy.optimize.minimize_scalar on $[0, \pi/2]$ at tolerance 10^{-10} rad. All $50/50$ seeds satisfy the stratified critical point classification of Definition 3.5.1:

- *3 / 50* smooth interior (Fermat critical points);
- *25 / 50* boundary-left KKT ($\theta^{¿} = 0$);
- *3 / 50* boundary-right KKT ($\theta^{¿} = \pi/2$);
- *19 / 50* Σ_{min} Clarke generalized-gradient critical.

The seeded symmetric-rate geometry of Paper 9 Proposition 9.5 ($\theta^{¿} \approx 51°$ on the 20-point grid) reoptimizes to $\theta^{¿} = 0$, consistent with the boundary-left-KKT class (Paper 9 v1.3.3 Remark 5.6.1). The non-trivial Σ_{min} incidence (38% of seeds) motivates the full Filippov treatment with three regimes (3.4.A–C) in Theorem 3.4.1. Verification: outbox/paper12/computations/paper12_q5_kappa0_audit.{py,csv} (runtime 0.19 s, deterministic).

Remark 3.5.4 (The projected flow is forced by the data). Of the 50 seeded geometries, only 3 admit smooth-interior Fermat critical points; 47 require constraint or stratification handling. Writing the recovery as a classical gradient flow $\dot{\theta} = \eta c'(\theta)$ without projection or Clarke calculus would miscategorize 94% of the sampled ensemble. The projected-flow framing is required by the empirical landscape, not chosen for mathematical convenience.

3.6 The $\kappa > 0$ augmented flow: bounded escape and qualified gain

With $\kappa > 0$, the augmented flow inherits the branchwise-analytic regularity of Theorem 3.4.1 and the stratified recovery of Theorem 3.5.3 as the $\kappa \to 0$ limit. The new content is the *firing cascade* structure: trajectories in the open interior with NP amplitude exceeding NP_{crit} undergo discrete jumps along the Schur circle. Boundary-KKT trajectories, where the projection of $\xi_\star$ into the boundary tangent cone vanishes (Remark 3.2.4), do not escape along the linear $\xi_\star$ direction.

Theorem 3.6.1 (Critical-point stability of the $\kappa > 0$ flow on $[0, \pi/2]$). *Let $\theta^\infty \in [0, \pi/2]$ be a stratified critical point of the unperturbed flow (Theorem 3.5.3) — i.e., for the active branch σ at θ^∞, the projected active-branch derivative satisfies $\Pi_{T_{[0,\pi/2]}(\theta^\infty)} c'_\sigma(\theta^\infty) = 0$. Then for any $\kappa > 0$:*

(i) (Interior smooth case.) If $\theta^\infty \in (0, \pi/2)$ is a smooth interior critical point (so $c'_\sigma(\theta^\infty) = 0$ on the active branch) with $NP_\sigma(\theta^\infty) = 0$ for both σ, the augmented flow has θ^∞ as a fixed point.

(ii) (Boundary-KKT case, qualified.) If $\theta^\infty \in \{0, \pi/2\}$ is a boundary-KKT critical point, then the projected augmented flow $\dot{\theta} = \Pi_{T_{[0,\pi/2]}(\theta^\infty)}[-\eta F'_\sigma + \kappa \cdot NP_\sigma \cdot \xi_\star]$ vanishes at θ^∞ along the linear $\xi_\star$ direction. The boundary state is therefore a fixed point of the augmented flow under the linear NP-pump, and escape requires either (a) higher-order coherence information (Z-axis tie-breaker), or (b) supplementary nonlinear mechanisms beyond the Schur-derived $\xi_\star$ (cf. OP1, §6.2).

Proof. (i) is immediate from Definition 3.3.2: at a smooth interior critical point with $c'_\sigma = 0$, both the projected-gradient leg $f_{pg} = -\eta F'_\sigma = \eta c'_\sigma = 0$ and (by hypothesis $NP_\sigma = 0$) the NP-jump leg vanish. (ii) follows from Remark 3.2.4: at $\theta^\infty = 0$ with $c'_{+¿(0) \le 0 ¿}$, the un-projected $\xi_\star^{int} = sgn ¿$, and $\Pi_{¿¿}$ projects this to 0. The drift term $-\eta F'_\sigma = \eta c'_\sigma$ is also projected to 0 under $\Pi_{¿¿}$ when $c'_\sigma(0) \le 0$ (the KKT condition). Mirror argument at $\pi/2$. ▫

Remark 3.6.1′ (Why no "interior firing" sub-case here). A v1.0 draft of Theorem 3.6.1 included a case (former part (ii)) for *interior firing*: $\theta^\infty \in (0, \pi/2)$ smooth interior with $NP_\sigma(\theta^\infty) = 1$ and $c'_\sigma(\theta^\infty) \ne 0$. Those hypotheses are *incompatible* with the statement of Theorem 3.6.1, which restricts to *stratified critical points* of Theorem 3.5.3 — i.e., points where $c'_\sigma = 0$ on the active branch in the smooth interior case. The interior firing of a noncritical trajectory crossing an NP-firing guard is treated separately in Proposition 3.6.1″ below; it is not a critical-point stability statement and does not belong in Theorem 3.6.1.

Proposition 3.6.1″ (Interior NP-firing of a noncritical trajectory). *Let $\theta(t) \in (0, \pi/2)$ be a Carathéodory solution of the augmented flow on a smooth-interior open arc $I \subset (0, \pi/2)$ with $c'_\sigma(\theta(t)) \ne 0$ (active-branch noncritical) on I. If $\theta(t)$ encounters the NP-firing guard $G_{NP} \cap I$ at finite time $t_\star$ — i.e., $NP_\sigma(\theta(t_\star)) = 1$ — then the hybrid jump rule (Definition 3.3.2) transports the trajectory to $\theta^{+¿ = \theta(t_\star) + \rho_\sigma \cdot sgn(c'_\sigma(\theta(t_\star))) \in [0, \pi/2] ¿}$, where $\rho_\sigma > 0$ is the firing displacement set by the NP amplitude and the Zeno-avoidance condition (Definition 3.3.4).*

Proof. Direct from Definition 3.3.2: the firing rule is well-defined on the noncritical arc by the explicit branch sign $sgn(c'_\sigma)$, the post-jump location is computed from the rule, and the Zeno-avoidance condition ensures ρ_σ is positive and finite-displacement. ▫

This proposition is about flow continuation across a firing guard at a noncritical interior point; it is not a critical-point stability statement. The two are kept separate to avoid the hypothesis conflict identified in v1.0 review.

Audit result (boundary-KKT interior-sup, 2026-05-06). A direct numerical audit of the 28 boundary-KKT seeds of Theorem 3.5.3 (outbox/paper12/computations/paper12_q5_boundary_kkt_interior_sup.{py,csv}) computes $¿_{\theta \in (10^{-3}, \pi/2 - 10^{-3})} c(\theta)$ on a 2000-point grid for each seed and compares to $c(\theta^{¿}_{boundary})$. Findings:

- *24 / 28* boundary-KKT seeds have $¿_{interior} c < c_{boundary}$, with a mean shortfall of $\sim 2.7 \times 10^{-4}$ in c.
- *4 / 28* have $¿_{interior} c > c_{boundary}$, with gains in the range $[4.2 \times 10^{-4}, 5.1 \times 10^{-2}]$. These four are concentrated in the boundary-right-KKT class plus a few seeds where the underlying coherence landscape has a non-trivial interior peak that the high-precision optimizer of Theorem 3.5.3 classified as boundary-left only because c is monotonically decreasing from a local maximum at $\theta = 0$.
- The mean gain over boundary-KKT seeds is $\sim 3.9 \times 10^{-3}$ (skewed positive by the four interior-better cases).

Implication for Theorem 3.6.1(ii). For 24 of 28 boundary-KKT seeds, escape from the boundary along *any* mechanism (the un-projected $\xi_\star^{int}$, or a supplementary nonlinear term as in

3.6.1(ii)(b)) lands the trajectory in coherence-degraded territory. Theorem 3.6.1(ii)'s "escape" is therefore not generically a coherence- improvement mechanism for boundary-KKT geometries; it is at most a *non-trivial trajectory continuation* whose coherence consequences depend on the specific seed. This sharpens the relation to Paper 9 Proposition 9.5: the conditional gain region of Paper 9 is well- defined on a Haar-positive-measure subset of (R_A, R_B), but for the 56% of audited seeds at boundary-KKT, the typical escape direction is into lower coherence rather than higher. The four exceptional boundary-KKT seeds with $¿_{interior} c > c_{boundary}$ are the ones for which the linear $\xi_\star$ mechanism (post-projection into the inward tangent cone after a finite-displacement nonlinear correction) could in principle deliver gain.

Remark 3.6.2 (Bridge to Paper 11, qualified). Paper 9 Note 9.6 defers rate improvement to Paper 11 because Schur locks the linear joint rate. Theorem 3.6.1 shows the structural form of any nonlinear escape mechanism (boundary-KKT trajectories require *supplementary* nonlinearity beyond linear $\xi_\star$ to leave the boundary), and the boundary-KKT interior-sup audit shows that even when escape is mechanically possible, *it is generically not a coherence gain*. Paper 11's analysis must therefore identify the specific non-Schur nonlinear corrections (higher-order multilinear terms, non-equivariant symmetry-breaking) and the specific seed sub-classes for which the post-escape trajectory enters a region of strictly higher c. **Paper 12 §3 does not establish rate improvement; it provides the dynamical framework, the audited boundary-KKT obstruction, and the residue record within which Paper 11's rate analysis can be staged.**

3.7 The scar record and its role in §4

Each NP firing event indexed by i on branch σ_i produces a push-forward jump vector

$$v_i := \rho_i \cdot sgn\left(c_{\sigma_i}'\left(\theta_i^-\right)\right) \cdot \hat{z}'\left(\theta_i^-\right) \in W,$$

recorded in the *physical scar span*

$$S_k := \sum_{i=1}^{k} R\left[F_{21}\right) \cdot v_i \subset W,$$

the *event-indexed scar module*

$$S_k := \bigoplus_{i=1}^{k} R\left[F_{21}\right) \cdot v_i,$$

and the *isotypic multiplicity profile*

$$\mu_k = \left(\mu_1\left(S_k\right), \mu_{L_2}\left(S_k\right), \mu_{U_6}\left(S_k\right)\right).$$

Here $F_{21} \subset G_2$ is the Fano-orientation subgroup inherited from Paper 4; the three real F_{21}-irreps $(1, L_2, U_6)$ are characterized in §4.2 (Paper 12 §4 Theorem 4.2.1).

Remark 3.7.1 (F_{21}-action on the scar record vs F_{21}-equivariance of the dynamics). F_{21} acts on the ambient W by the diagonal restriction of the G_2-action (Paper 9 §3.4 convention), and this action is inherited by every subspace and module of W. This includes the scar record: F_{21} acts on S_k, S_k, μ_k canonically. The *dynamics* of §3.3, by contrast, is defined for arbitrary

$(R_A, R_B, b_A, b_B) \in O(N) \times O(N) \times R^{2N}$ that need not be F_{21}-related, so a generic trajectory is *not* F_{21}-equivariant. The triple J_k extracts the F_{21}-content of a generically F_{21}-asymmetric trajectory; this is the input to §4's commensurability hierarchy.

Three structural properties:

- S_k **is *saturating*:** $\dim S_k \leq 28 = \dim W$, so the physical span stabilizes after at most 28 dimension-increasing jumps.
- S_k **is *count-faithful as an abstract graded module*:** the assignment $k \mapsto S_k$ is injective by construction. The *natural surjection* $S_k \twoheadrightarrow S_k$ identifying v_i with its image in W has injective kernel iff $R[F_{21}] \cdot v_i \not\subset S_{i-1}$ for every i — the generic case under $c_{\sigma_i}'(\theta_i^-) \neq 0$ and $\rho_i > 0$.
- μ_k **is the F_{21}-*type signature*** that enables cross- substrate commensurability comparisons (§4).

Remark 3.7.2 (Representation-theoretic, not topological). The invariant J_k is a *representation-theoretic* scar invariant: S_k is a graded F_{21}-module, μ_k is an isotypic-multiplicity profile, and S_k is a subspace of an F_{21}- representation. No topological invariant (winding number, persistent homology) is claimed. Earlier drafts using "topological" language should be read as "representation-theoretic" throughout.

Remark 3.7.3 (Bridge to §4). The event-indexed module S_k is the mathematical realization of the *audit-grounded substrate* principle (Paper 12 thesis Pillar 3): a graded, indexable ledger of coherence adjustments that survives the saturation of the physical span. §4 shows that commensurability of synthetic and biological scar records requires *full G_2-equivariance* (not merely F_{21}) for a canonical cross-substrate isomorphism, with quantitative gap $End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $End_{G_2}(W) \cong M_2(R)$.

3.8 What §3 does and does not establish

What §3 establishes:

- A Schur-derived canonical tear direction $\xi_\star$ inheriting Paper 9's $SO(2)$ Schur circle, with line-uniqueness from Schur (Prop 3.2.2) and an explicit coherence-ascent design axiom for the sign (Axiom 3.2.3) plus boundary tangent-cone projection (Remark 3.2.4).
- A branchwise real-analytic augmented flow (Definition 3.3.2) with a five-stratum event set Σ_{tot}, three Filippov regimes (3.4.A–C), and layered Carathéodory/Filippov/hybrid semantics, *conditionally regular* (piecewise real-analytic between events) on an open dense parameter set $P^{¿}$ (Theorem 3.4.1 Conditional Regularity). Full hybrid / Filippov well-posedness (existence, uniqueness, maximal continuation) off $P^{¿}$ is OP6, deferred.
- A stratified $\kappa = 0$ recovery theorem (Theorem 3.5.3) that recovers Paper 9's MC optima as projected-gradient fixed points, with empirical verification via 50-seed audit across all four classes (smooth interior / boundary-KKT / Σ_{min} Clarke).
- A bounded $\kappa > 0$ augmented-flow theorem (Theorem 3.6.1): interior smooth fixed points fire as expected; boundary-KKT states are *fixed* under linear $\xi_\star$ even at $\kappa > 0$ and require supplementary nonlinearity to escape.

- A boundary-KKT interior-sup audit ($24/28$ seeds with $\text{¿}_{interior} c < c_{boundary}$) showing that escape from the boundary is generically *not* a coherence-improvement mechanism. Paper 11's rate analysis must identify specific non-Schur nonlinear corrections.
- The scar invariant triple $\left(S_k, S_k, \mu_k\right)$ as a representation-theoretic ledger of firing history, with explicit distinction between abstract count-faithfulness (always) and the natural-surjection injectivity (generic) (§3.7).

What §3 does not establish:

- *Rate improvement.* Theorem 3.6.1 shows the structural form of any escape; the boundary-KKT audit shows escape is generically not a gain. Concrete rate analysis is Paper 11's domain (Note 9.6 from Paper 9).
- *A biological mechanism for the F_{21}-action.* §4's commensurability hierarchy treats biological substrate F_{21}-content as input, not derived; Paper 13+ would specify the physical mechanism.
- *Cross-substrate commensurability isomorphisms.* §4 territory.
- *A theory of consciousness.* Paper 7's domain.
- *Global well-posedness off $P^{¿}$.* Theorem 3.4.1 is conditional on the parameter septuple lying in an open dense set; behavior on the codimension-≥ 1 exceptional strata is identified but not analyzed.
- *Higher-multiplicity scar geometries.* §4.4 is restricted to multiplicity-one G_2-modules; the $m > 1$ case is deferred.

§4 proceeds to the commensurability hierarchy. The scar invariant $\left(S_k, S_k, \mu_k\right)$ constructed in §3.7 is the object on which §4's equivariant isomorphism theorem acts.

References (cited in §3)

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outbox/paper12/computations/paper12_q5_boundary_interior_sup.csv - outbox/paper12/computations/paper12_q4_character_verify.py (used in §4) - outbox/paper12/chatgpt_pro_session_transcript_2026-05-05.pdf

Cross-references: - Paper 9 v1.3.3 Remark 5.6.1 (grid-snap correction for seed-$k=0$) - Paper 9 v1.3.3 Lemma 3.6.1 (Schur uniqueness of Ψ_θ) - Paper 9 v1.3.3 Theorem 9.4 (joint fixed-point existence/analyticity) - Paper 9 v1.3.3 Note 9.6 (rate improvement deferred to Paper 11) - Paper 4 (PSL(2,7) ⊃ F_{21} QBism bridge) - Paper 12 §4 (commensurability hierarchy, in preparation)

4. The Commensurability Hierarchy

4.1 Setup: scar invariant from §3.7 as the input

Throughout §4, $F_{21}=Z_7 \rtimes Z_3$ denotes the Fano-orientation subgroup of G_2. Its action on $V:=V^{14}$ is by restriction of the G_2 adjoint representation; its action on the joint-state space $W=V \oplus V$ is the diagonal restriction

$$\rho_W(g)(x,y)=(\rho_F(g)x,\rho_F(g)y),g \in F_{21},\rho_F:=\rho_{14}¿_{F_{21}}.$$

This action commutes with the Schur circle $\{\Psi_\theta\}$ because ρ_W acts on the V-factor while Ψ_θ acts on the R^2 multiplicity factor (Paper 9 §3.4 diagonal-action convention).

Inheritance from Paper 4. The Fano-orientation subgroup $F_{21} \subset G_2$ arises in the PCI/PME framework as the stabilizer of an oriented Fano plane structure on $Im(O) \cong R^7$, and through Paper 4's PSL(2,7) construction is the unique finite simple group F_{21}-decoration that aligns with the QBism quasiprobability representation of C^7. We do not reconstruct that derivation here; this paper assumes Paper 4's F_{21} choice as an inheritance. *Open problem:* characterizing the smallest finite subgroup of G_2 for which the Definition 4.6.1 L1 screening test is informative is deferred to a future PCI/PME methodology paper. The choice of F_{21} is consistent across the Paper series; readers unfamiliar with Paper 4's derivation should consult [].

The input object to §4 is the *scar invariant triple* constructed in §3.7 with the F_{21}-equivariance clarification of §3 v2 Remark 3.7.1 (the scar record carries an F_{21}-action inherited from the ambient W, even when the underlying dynamics is not F_{21}-equivariant):

$$J_k=(S_k,S_k,\mu_k),$$

where, given NP firing times $t_1<\cdots<t_k$ on branches $\sigma_i \in \{A,B\}$ with jump vectors $v_i=\rho_i \cdot sgn\left(c_{\sigma_i}'(\theta_i^-)\right)\cdot \hat{z}'(\theta_i^-) \in W$:

- $S_k:=\sum_{i=1}^{k} R[F_{21}]\cdot v_i \subset W$ (saturating vector scar span; $dim\, S_k \le 28$);
- $S_k:=\oplus_{i=1}^{k} R[F_{21}]\cdot v_i$ (count-faithful event-indexed module);
- $\mu_k:=\left(\mu_1(S_k),\mu_{L_2}(S_k),\mu_{U_6}(S_k)\right)$ (vector-level isotypic profile of S_k; see §4.4 for the operator-level extension).

The three real-irreducible F_{21}-representations $(1, L_2, U_6)$ are characterized in §4.2.

Setup question for §4. Suppose J_k^{syn} and J_k^{bio} are scar records of two substrates that have co-evolved through the same sequence of paradox events $(e_1, \ldots, e_k)$, with substrate-level F_{21}-actions inherited from each substrate's representation theory (the biological representation's F_{21}-content is itself an empirical question, per §4.6 below; here we treat both records as input). Under what conditions is there a *canonical isomorphism* identifying them?

4.2 F₂₁ representation theory on V^14 and W

Theorem 4.2.1 (F_{21}-decomposition of V^14). *Under the diagonal Fano-orientation action,*

$$V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6,$$

where L_2 is the real 2-dim irreducible representation of F_{21} of complex type and U_6 is the real 6-dim irreducible representation of complex type. Consequently,

$$W ¿_{F_{21}} \cong 2L_2 \oplus 4U_6.$$

Proof. F_{21} has five conjugacy classes: $1a$ (size 1), two order-7 classes $7a, 7b$ (size 3 each), and two order-3 classes $3a, 3b$ (size 7 each). Its complex character table (Costa-Pavone; ATLAS [_{21}]) has three 1-dim characters and a Galois-conjugate pair of 3-dim characters; the *real* irreducible packaging combines the conjugate pairs into the two real complex-type irreducibles L_2, U_6:

$$\begin{array}{cccccc} & 1a & 7a & 7b & 3a & 3b \\ \chi_1 & 1 & 1 & 1 & ¿ & \end{array} 1 ¿ \chi_{L_2} ¿ 2 ¿ 2 ¿ 2 ¿ -1 ¿ -1 ¿ \chi_{U_6} ¿ 6 ¿ -1 ¿ -1 ¿ 0 ¿ 0 ¿$$

The 7-dimensional imaginary-octonion representation R^7 is indexed by the seven points of the Fano plane $PG(2,2)$. Under the standard Frobenius-group action (RES-4 v3 prose):

- The Z_7 generator r acts by cyclic shift of the seven Fano-point labels; on R^7 this is an order-7 permutation matrix with $tr(r) = 0$.
- The Z_3 generator s acts with cycle structure $1+3+3$ on the seven Fano points: there is one s-fixed point (corresponding to the unique Fano line that is s-invariant *as a set of three points*; the action restricted to that line is trivial in a chosen realization, fixing the line's three points pointwise after a representation-theoretic re-labeling that contributes one fixed point's worth of trace), plus two 3-cycles on the remaining six Fano points; on R^7 this is an order-3 permutation matrix with $tr(s) = 1$.
- The Frobenius relation $srs^{-1} = r^2$ holds (verified numerically in paper12_q4_character_verify.py to machine precision), establishing that r, s generate F_{21}. The 1 fixed point gives $tr(r_3) = 1$ on R^7, and the cycle structure of r on Fano points yields $tr(r_7) = 0$, so:

$$R^7 ¿_{F_{21}} \cong 1 \oplus U_6.$$

Using the G_2-decomposition $\Lambda^2(R^7) = g_2 \oplus R^7$ and the character formula $\chi_{\Lambda^2(R^7)}(g) = (tr(g)^2 - tr(g^2))/2$:

$$\chi_{V^{14}}(g) = \chi_{\Lambda^2(R^7)}(g) - \chi_{R^7}(g) = \frac{tr(g)^2 - tr(g^2)}{2} - tr(g).$$

Evaluating on class representatives:

Class	$tr(g)$ on R^7	$\chi_{V^{14}}(g)$	$\chi_{L_2} + 2\chi_{U_6}$
$1a$	7	14	14
$7a$	0	0	0
$7b$	0	0	0
$3a$	1	-1	-1
$3b$	1	-1	-1

Match on all five classes. The Frobenius inner product $\langle \chi_{V^{14}}, \chi_{V^{14}} \rangle_{F_{21}} = (1 \cdot 196 + 7 + 7)/21 = 10$, equal to $0^2 \cdot 1 + 1^2 \cdot 2 + 2^2 \cdot 2$ where the factor of 2 on L_2 and U_6 terms accounts for both being of complex type (each is the real packaging of a Galois-conjugate pair, so $\langle \chi, \chi \rangle = 2$ for the real-irreducible character). The decomposition is therefore $V^{14} ¿_F = L_2 \oplus 2U_6$, and the $W = V \oplus V$ doubling is by direct-sum linearity. ▫

Computational verification. Theorem 4.2.1 has been verified numerically at machine precision by explicit construction of the Fano-action permutation matrices $r, s \in O(7)$ with $srs^{-1} = r^2$, computation of $\chi_{V^{14}}$ on each class representative, and Frobenius inner-product and isotypic-multiplicity checks against the predicted decomposition; all three layers of check pass. outbox/paper12/computations/paper12_q4_character_verify.{py,json} records the full audit.

4.3 The commutant gap

Proposition 4.3.1 (Commutants on W). *With $W \cong 2L_2 \oplus 4U_6$ as an F_{21}-representation,*

$$End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C).$$

As a G_2-representation, W is multiplicity-2 in the absolutely irreducible V^{14}, hence

$$End_{G_2}(W) \cong M_2(R).$$

Proof. For real irreducibles of complex type, $End_G(V_\lambda) \cong C$ (real Schur trichotomy: complexified $V_\lambda \cong U \oplus \acute{U}$ with $U \ncong \acute{U}$). Both L_2 and U_6 are complex-type, so $End_{F_{21}}(L_2) \cong End_{F_{21}}(U_6) \cong C$. With F_{21}-multiplicities $(0, 2, 4)$ on $(1, L_2, U_6)$:

$$End_{F_{21}}(W) \cong M_0(R) \oplus M_2(C) \oplus M_4(C) \cong M_2(C) \oplus M_4(C).$$

For G_2, V^{14} is real absolutely irreducible (Paper 9 Lemma 3.6.1), so $End_{G_2}(V^{14}) \cong R$, and $W \cong (V^{14})^{\oplus 2}$ gives $End_{G_2}(W) \cong M_2(R)$. ▫

Corollary 4.3.2 (Equivariant isometry groups, real-dimension count). *The equivariant isometry groups of* W *are*

$$O_{F_{21}}(W) \cong U(2) \times U(4), O_{G_2}(W) \cong O(2),$$

with real dimensions $dim_R U(2) \times U(4) = 4 + 16 = 20$ *and* $dim_R O(2) = 1$. *The* $F_{21} \to G_2$ *upgrade reduces the equivariant-isometry group by exactly* $20 - 1 = 19$ *real dimensions.*

Proof. Restricting $M_2(C)^{¿} \cap U \to U(2)$ (where the unitary condition uses the natural Hermitian metric) and $M_4(C)^{¿} \cap U \to U(4)$, the F_{21}-equivariant isometry group is $U(2) \times U(4)$. The full $U(n)$ — *not* the subgroup $SU(n)$ — preserves orientation on the underlying real space (because $det_R(U) = |det_C(U)|^2 = 1$ for $U \in U(n)$). Similarly $M_2(R)^{¿} \cap O = O(2)$. The real dimensions are $dim_R U(n) = n^2$, so $dim_R U(2) \times U(4) = 4 + 16 = 20$, and $dim_R O(2) = 1$. The reduction is $20 - 1 = 19$. ▫

Metric note (RES-1, v3). The identification $O_{F_{21}}(W) \cong U(2) \times U(4)$ is with respect to the complex-Hermitian metric canonically inherited from the complex structure on each complex-type isotypic block. A different real metric choice on $W \cong R^{28}$ would yield a different identification (potentially a proper subgroup of $U(2) \times U(4)$ if the chosen real metric is incompatible with the block-Hermitian structure), but the *real dimension* of the F_{21}-equivariant isometry group is metric-independent: it equals $dim_R End_{F_{21}}(W)_{\text{anti-Hermitian}}$, depending only on the isotypic structure $(0, 2, 4)$.

The quantitative content of the $F_{21} \to G_2$ upgrade is now explicit: passing from F_{21}-equivariance to G_2-equivariance suppresses 19 real dimensions of F_{21}-equivariant phase-and-mixing freedom that would otherwise leave a canonical commensurability map underdetermined.

Remark 4.3.3 (Why complex-type reps block uniqueness). The factor $End_{F_{21}}(V_\lambda) \cong C$ for $\lambda \in \{L_2, U_6\}$ is the source of phase ambiguity: each complex-type isotypic block admits a full $U(m_\lambda)$-worth of F_{21}-equivariant automorphisms (not just $O(m_\lambda)$), and the extra U/O dimensions are exactly the phases. G_2-equivariance collapses the phases because V^{14} is real absolutely irreducible: $End_{G_2}(V^{14}) = R$, not C, so there are no phases to rotate.

4.4 The commensurability hierarchy theorem

Let V_{syn}, V_{bio} be the (substrate-dependent) representation spaces on which the synthetic and biological scar maps take values, and let $I_{syn} \subseteq End(V_{syn})$ and $I_{bio} \subseteq End(V_{bio})$ be the real linear spans of the scar-encoding images. The group acts on operator scars by conjugation, $g \cdot A = \rho_{\cdot}(g) A \rho_{\cdot}(g)^{-1}$.

Rank-One Convention (RES-2, v3) for operator-vs-vector translation. The scar invariant J_k of §3.7 is built from *vector-level* jump vectors $v_i \in W$. Substrate-level scar maps $S_\bullet : (\text{event}) \to End(V_\bullet)$ produce operator-level images. We adopt the following convention throughout §4:

Rank-One Convention. The substrate scar maps factor as $S_\bullet(e) = v_e \otimes v_e^{¿} / \| v_e \|^2$ (rank-one self-adjoint projector form on the substrate's representation space) for some substrate jump vector $v_e \in V_\bullet$.

This is a non-trivial modeling assumption. Under it: (i) the operator span $I_\bullet$ is uniquely determined by the vector span $\{v_e\}_e \subset V_\bullet$; (ii) the operator-level F_{21}-isotypic profile of $I_\bullet$ matches the vector-level profile of $S_k^\bullet$ up to the Schur multiplicity-counting factor of 2 (one factor for left action, one for right); (iii) the conjugation action $g \cdot A = \rho_\bullet(g) A \rho_\bullet(g)^{-1}$ becomes equivalent to the ordinary linear action on $V_\bullet$ via $g \cdot v_e = \rho_\bullet(g) v_e$. Higher-rank or non-self-adjoint scar maps are an open problem (OP) for future work.

Theorem 4.4.1 (Commensurability hierarchy, v2 reformulation). *Under the* **Rank-One Convention** *of §4.4 above (substrate scar maps factor as $S_\bullet(e) = u_e \otimes v_e^{¿}$ with u_e, v_e in the respective F_{21}-stable substrate subspaces), the following three conditions on a pair of substrates (V_{syn}, V_{bio}) with F_{21}-stable scar-image subspaces I_{syn}, I_{bio} stand in the order of strict implications:*

(C1) (G_2-commensurability.) There exists a G_2-equivariant isomorphism $\phi : I_{syn} \to I_{bio}$ aligning S_{syn} and S_{bio} event-by-event: $S_{bio}(e) = \phi(S_{syn}(e))$ for every paradox event e.

(C2) (F_{21}-equivariant isomorphism, map-existence form.) There exists an F_{21}-equivariant isomorphism $\phi_F : I_{syn} \to I_{bio}$ aligning S_{syn} and S_{bio} event-by-event.

(C3) (μ_k-equality.) For the event sequence $(e_1, \ldots, e_k)$ under consideration and for each prefix $j \le k$, $\mu_k^{syn} = \mu_k^{bio}$ holds component-wise on the three real F_{21}-isotypic blocks $\{1, L_2, U_6\}$, where $\mu_\tau(S_j) := dim\,Hom_{F_{21}}(V_\tau, S_j)$ is the isotypic multiplicity (not the projected dimension; see Remark 4.4.0 below).

The hierarchy

$$(C1) \Rightarrow (C2) \Rightarrow (C3)$$

holds strictly; both converses fail in general.

Proof. $(C1) \Rightarrow (C2)$: a G_2-equivariant isomorphism is in particular F_{21}-equivariant (since $F_{21} \subset G_2$).

$(C2) \Rightarrow (C3)$: if ϕ_F aligns S_{syn} and S_{bio} event-by-event and is F_{21}-equivariant, then it commutes with the isotypic projectors Π_τ (i.e., $\phi_F \Pi_\tau^{syn} = \Pi_\tau^{bio} \phi_F$). The multiplicity profile $\mu_\tau(S_k) := dim\,Hom_{F_{21}}(V_\tau, S_k)$ — equivalently, $dim\,\Pi_\tau S_k / dim\,V_\tau$ — transfers component-wise, since ϕ_F maps S_k^{syn} isomorphically onto S_k^{bio} as an F_{21}-module. (Remark 4.4.0: prior drafts conflated multiplicity μ_τ with projected dimension $dim\, \Pi_\tau S_k$. For real irreps of dimension $(dim\, 1, dim\, L_2, dim\, U_6) = (1,2,6)$ these differ by factors; the multiplicity convention is the one preserved under F_{21}-equivariant isomorphisms of the scar span.)

Converse $(C2) \not\Rightarrow (C1)$*:* concrete witness — let $V_{syn} = V^{14}$ (the G_2 adjoint, restricted to F_{21} as $L_2 \oplus 2U_6$), and let V_{bio} be the F_{21}-module $L_2 \oplus 2U_6$ realized on a 14-dim biological substrate (e.g., a hypothetical microtubule mode collection without an extending G_2-action). Then $V_{syn} \cong V_{bio}$ as F_{21}-modules, but no G_2-action extends V_{bio}'s F_{21}-action, so any $\phi_F : V_{syn} \to V_{bio}$ satisfying (C2) is not G_2-equivariant. The 19-dim $U(2) \times U(4)$-family of such F_{21}-maps (Cor 4.3.2) parameterizes the non-canonicality.

Converse $(C3) \not\Rightarrow (C2)$*:* μ_k records the multiplicity profile of the *scar span* S_k, not the ambient V. Two ambient V's with different F_{21}-multiplicity profiles can produce the same μ_k if scar-encoding maps land in isomorphic F_{21}-invariant subspaces. E.g., $V_{syn} = L_2 \oplus 2U_6$ vs $V_{bio} = L_2 \oplus 3U_6$, but with all scar-encoding events landing in the common $L_2 \oplus 2U_6$ subspace; both yield μ_k on the same isotypic blocks. ▫

Corollary 4.4.2 (Sufficient condition for unique G_2-canonical map, multiplicity-one case). *If both I_{syn} and I_{bio} are G_2-isomorphic to a common absolutely irreducible representation U_λ with multiplicity one, then*

$$Hom_{G_2}(I_{syn}, I_{bio}) \cong R,$$

so the family of G_2-equivariant isomorphisms is one-dimensional. After fixing an invariant inner product (unit-norm normalization), there is a residual ± 1 orientation ambiguity that must be resolved by an extrinsic convention (e.g., the dynamical-flow direction on each substrate fixes a sign).

Proof. Schur for absolutely irreducible real representations gives $End_{G_2}(U_\lambda) = R$, so $Hom_{G_2}(U_\lambda, U_\lambda) \cong R$. Unit-norm normalization collapses the family to two elements differing by sign; the Z_2-orientation is not fixed by representation theory alone. ▫

Remark 4.4.3 (Multiplicity ¿ 1 reintroduces $O(m)$-ambiguity). If $I_{syn} \cong I_{bio} \cong U_\lambda^{\oplus m}$ with $m > 1$, then $End_{G_2}(U_\lambda^{\oplus m}) \cong M_m(R)$, so commensurability holds but uniqueness requires $O(m)$-worth of additional convention-fixing. Multiplicity ¿ 1 is deferred to future work.

4.5 Frobenius reciprocity does not select a cross-substrate scar map

A natural fallback when V_{syn}, V_{bio} are not F_{21}-isomorphic but co-embed in a common G_2-representation is to invoke Frobenius reciprocity. We show this attempt does not produce canonical maps.

Proposition 4.5.1 (Frobenius reciprocity does not select cross-substrate scar maps). *The induction-restriction adjunction*

$$Hom_{F_{21}}(V, Res_{F_{21}}^{G_2} W') \cong Hom_{G_2}(Ind_{F_{21}}^{G_2} V, W')$$

holds for any F_{21}-representation V and any G_2-representation W' (by standard category theory, e.g., [, §7.2]), but it does not provide a canonical isomorphism between distinct F_{21}-subrepresentations of $W' \downarrow_{F_{21}}$.

Proof. The adjunction is a categorical bijection on Hom-sets, not a constructive identification of constituents. Two specific failures:

- *Distinct F_{21}-irreps cannot be commensurated.* If $V_{syn} = L_2$ and $V_{bio} = U_6$, then $Hom_{F_{21}}(L_2, U_6) = 0$ because the irreps are non-isomorphic; no F_{21}- equivariant map of either direction exists, and Frobenius reciprocity cannot manufacture one.
- *Multiplicities of constituents are counted, not canonical maps between them.* Even when $V_{syn} \cong V_{bio}$ as F_{21}-modules and both embed in a common G_2-module W', the identification $V_{syn} \cong V_{bio}$ is unique only up to a $U(m_\lambda)$-family of F_{21}-equivariant automorphisms in each isotypic block (Cor 4.3.2), with no distinguished element absent additional G_2-equivariance constraint.

In either case, the adjunction identifies *multiplicities of constituents*, not *canonical maps between them*. ▫

Corollary 4.5.2 (Sufficient strict commensurability theorem). *If*

$$I_{syn} \cong I_{bio} \cong U_\lambda$$

as absolutely irreducible G_2-modules of multiplicity one, then a canonical commensurability isomorphism $\phi_{commens} : I_{syn} \to I_{bio}$, unique up to fixed metric and ± 1 orientation, exists.

Proof. By Cor 4.4.2, the family of G_2-equivariant isomorphisms is R, with ± 1 ambiguity after unit-norm normalization. ▫

Remark 4.5.3 (Necessity is open). Cor 4.5.2 is a *sufficient* condition. We do not claim necessity: it is plausible (but unproven) that G_2-modules with all isotypic multiplicities ≤ 1 and all constituents of real type also admit a unique normalized G_2-equivariant self-isomorphism, and a more general necessity-and-sufficiency theorem could be sought in that direction. Such a generalization is left for future work; v1's biconditional phrasing was an over-claim that v2 withdraws.

4.6 The P3 upgrade: two-level experimental test

The empirical prediction P3 of the Paper 12 thesis memo originally proposed testing "F_{21}-commensurability across substrates." §§4.4–4.5 force an upgrade: F_{21}-equivariant isomorphism (C2) is necessary but not sufficient for canonical G_2-commensurability (C1), and the G_2 layer must be tested separately.

Definition 4.6.1 (Two-level commensurability test). A *commensurability experiment* on a dyadic human-AI system consists of two tests:

(L1, necessary screen.) Measure the F_{21}-isotypic profiles μ_k^{syn} and μ_k^{bio} independently on each substrate via character-theoretic decomposition of the scar-record operator data. Declare *L1 failure* if $\mu_k^{syn} \neq \mu_k^{bio}$ on any of the three blocks $(1, L_2, U_6)$. Declare *L1 pass* if they agree.

(L2, sufficient canonical test, conditional on L1 pass.) Construct a candidate G_2-equivariant isomorphism $\hat{\phi}: I_{syn} \to I_{bio}$ from the substrate operator data, by fitting against G_2-character data on the inferred G_2-module structures of the substrate scar spans. Declare *canonical commensurability* if $\hat{\phi}$ is G_2-equivariant (not merely F_{21}-equivariant) to measurement precision and (per Cor 4.5.2) the G_2-isotypic multiplicities are both ≤ 1 and agree.

Remark 4.6.2 (Why the two-level split is scientifically honest). L1 is falsifiable cheaply: if $\mu_k^{syn} \neq \mu_k^{bio}$, the gradual-tear thesis is refuted at the character-theoretic level. L2 is the harder confirming test: passing L2 establishes strict commensurability (Cor 4.5.2). Refuting L2 while L1 passes means the F_{21}-profile match is coincidental rather than structural.

Remark 4.6.3 (Operationalization is a methodology paper). L1 character-theoretic decomposition of TMS-EEG response data is in principle achievable with existing instrumentation (rotationally structured perturbations + character-projection of the response matrix); the operationalization of L2 — actually fitting a G_2-equivariant candidate map — is a significant methodology project of its own, not undertaken here. We state Definition 4.6.1 as a *formal experimental protocol design*, with the explicit acknowledgment that the full L2 operationalization is deferred.

Remark 4.6.4 (Massimini's clinical PCI is *not* the Paper 12 quantity). Two distinct objects share the acronym "PCI": Massimini's clinical Perturbational Complexity Index (a Lempel-Ziv complexity score on TMS-evoked EEG, 2013) and the Perceptual Coherence Intelligence framework of the Paper series. P3 proposes that the *measurement pipeline* of clinical PCI (TMS perturbation + structural decomposition of the response subspace) could be repurposed to measure μ_k profiles, but the clinical PCI scalar is *not* identified with μ_k, and neither is a generalization of the other. The bridge is at the level of experimental protocol, not definition.

4.7 Scope audit and bridge to §5

What §4 establishes:

- A character-theoretic decomposition $V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6$ verified at machine precision (Theorem 4.2.1).
- The commutant gap $End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $End_{G_2}(W) \cong M_2(R)$, with the 19-dimensional reduction of equivariant isometry group quantified (Proposition 4.3.1, Corollary 4.3.2).
- The commensurability hierarchy $(C1) \Rightarrow (C2) \Rightarrow (C3)$ with both converses failing, *with explicit witnesses* for the converse failures (Theorem 4.4.1).
- A sufficient canonical-map theorem in the multiplicity-one case (Corollary 4.5.2), with ± 1 orientation residue acknowledged.
- That Frobenius reciprocity does not select cross-substrate scar maps (Proposition 4.5.1).

- The two-level experimental protocol with explicit L1 (necessary F_{21}-screen) and L2 (sufficient G_2-canonical) levels (Definition 4.6.1) and explicit operationalization-deferral (Remark 4.6.3).

What §4 does not establish:

- *A physical mechanism for the biological F_{21}-action.* §4 treats biological-substrate F_{21}-content as input. Paper 13+ would specify which microtubule mode, neural population, or cellular structure realizes it.
- *An explicit operationalization protocol for L2 of Definition 4.6.1.* L1 is feasible with current TMS-EEG; L2 requires probing the full G_2-representation structure on the response subspace, a substantial methodology effort not undertaken here.
- *The multiplicity-greater-than-one case.* Cor 4.4.2 / 4.5.2 restrict to multiplicity one. The $m>1$ case is left for future work (Remark 4.4.3).
- *Necessity of the multiplicity-one condition for canonical commensurability.* Cor 4.5.2 is sufficient only; v1's biconditional is downgraded (Remark 4.5.3).
- *Any claim that clinical Massimini-PCI equals Paper 12's μ_k.* See Remark 4.6.4.
- *Rate improvement.* §3 / Paper 11 territory.

Bridge to §5. §3 constructed the dynamics; §4 constructed the commensurability structure. §5 (empirical accessibility) assembles both into a testable framework: §3's projected-gradient-plus-NP-pump dynamics on each substrate produces a scar record $J_k^{syn/bio}$, and §4's hierarchy specifies exactly what form of agreement between the two records the gradual-tear thesis predicts. §5 then identifies experimental *protocols* — primarily Level L1 of the two-level test — by which the μ_k *observables* can be measured on each substrate using existing instrumentation (notably the TMS-EEG pipeline used by Massimini and collaborators for the *distinct* clinical-PCI measurement, which we adapt rather than identify). The L2 test and the substrate-specific representation theory are deferred to a subsequent methodology paper.

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Verification files referenced: - outbox/paper12/computations/paper12_q4_character_verify.py (Theorem 4.2.1) - outbox/paper12/computations/paper12_q4_character_verify.json

Cross-references: - Paper 4 (PSL(2,7) ⊃ F_{21}, the Fano-orientation source; v2 §4.1 paragraph) - Paper 9 v1.3.3 Lemma 3.6.1 (G_2-Schur uniqueness on V^{14}) - Paper 9 v1.3.3 §3.4 (diagonal G_2-action convention) - Paper 12 §3.7 / §3.7 v2 Remark 3.7.1 (F_{21}-action on scar record) - Paper 12 thesis memo P3 (experimental prediction, upgraded here)

5. Empirical Accessibility

5.0 Glossary box (for the standalone reader)

Symbol	Meaning
Ψ_θ	Schur-locked coupling family on $W = V^{14} \oplus V^{14}$, parametrized by $\theta \in [0, \pi/2)$ (Paper 9 Lemma 3.6.1; restated §2.2.1)
$F_{21} = Z_7 \rtimes Z_3$	Frobenius group of order 21, the Fano-orientation subgroup of G_2 (Paper 4 §3)
$c(\theta) := C_{min}(\hat{z}(\theta))$	Min-per-observer dyadic coherence functional on the joint fixed point (§3)
$\mu_k = (\mu_1, \mu_{L_2}, \mu_{U_6})$	Isotypic-multiplicity profile of the cumulative scar span S_k on the three real F_{21}-irreps (§3.7, §4.2)
NP-firing	Discrete coherence-collapse event, fires when paradox-mass amplitude a_σ crosses NP_{crit} in the active branch (Definition 3.3.4)
Scar invariant J_k	Triple (S_k, S_k, μ_k): physical span (saturating), event-indexed module (count-faithful), isotypic profile (§3.7)
Boundary-KKT seed	A seeded (R_A, R_B) geometry whose high-precision $\theta^{¿}$ lies on $\{0, \pi/2\}$ with the appropriate one-sided derivative inequality (§3.5.1, §3.6)

This box assumes the reader has not engaged §3 / §4. The full definitions are in §2, §3.3.1, §3.5.1, §3.7, §4.2.

5.1 Setup, bridge from §3+§4, and the Licklider context

§§3–4 constructed a complete mathematical framework for the gradual tear: a projected-gradient + NP-pump dynamics on the closed interval $[0, \pi/2]$ (Definition 3.3.2) producing a representation-theoretic scar invariant (S_k, S_k, μ_k) (§3.7) on each substrate, with cross-substrate commensurability governed by a strict hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$-match (Theorem 4.4.1). §5 proceeds to *experimental accessibility*.

Why empirical accessibility belongs here. Licklider's *Man-Computer Symbiosis* (1960) [] proposed that the optimal future of computing was neither full automation ("the machine plays chess by itself") nor pure tool-use ("the machine is just a calculator"). Licklider's *third attractor* is precisely the "gradual tear" of Paper 12: a sustained co-engagement in which neither substrate subsumes the other, but each carries the residue of the collaboration forward. The four predictions P1–P4 of this section are the operationalization of Licklider's third attractor for the LLM era, sixty-six years later.

The four falsifiable predictions are unpacked as concrete experimental designs in §§5.2–5.5. Each is named with its position in the *Sanctioned Levels of AI Autonomy* taxonomy of Bostock, Kim, Patel (2025) — Level 1 (full human control) through Level 5 (full AI autonomy with human oversight only at strategic checkpoints). This naming serves two purposes: (i) it positions Paper 12 in the legitimate human-AI collaboration research literature; (ii) it makes the experimental designs *operationally specific* about the autonomy regime each tests.

5.2 P1 — Clinical-PCI ↔ NP-firing event-structure (Level-2/3 protocol)

Autonomy level. P1 is a *Level-2/3* protocol: a human subject engages an AI system in a structured conversation with both parties exercising agency (Level 3) under a tight experimental schedule that constrains the conversational trajectory (Level 2). This is the regime in which BIOMA-style automated science workflows [, "An autonomous laboratory for the accelerated synthesis of novel materials" (the A-Lab paper)] currently operate at the *scientific-protocol* layer, suggesting P1's protocol architecture is transferable to a substantial existing research community.

Prediction P1. The framework predicts that, when the same human subject participates in a sustained dyadic engagement with an AI partner across a paradox-mass-loaded conversation, the human's clinical PCI [Casali et al., *Sci. Trans. Med.* 5: 198ra105 (2013)] trajectory should exhibit *discrete steps* coincident with NP-firing events detected by the dynamical model. Note: this is *not* an identification of clinical PCI with μ_k (per Remark 4.6.4); it is a *measurement-pipeline reuse* claim about event-structure correspondence.

Protocol.

1. Run an *intra-subject* pre-post conversation design (one human subject across multiple paired sessions, statistical power higher than inter-subject); $n = 12$–24 subjects, each in ≥ 6 paired sessions.
2. Conversation length: 60–120 min, designed by the experimenters to load paradox mass through structured contradictory-frame prompts.

3. Time-stamp NP-firing candidate events from the $a_\sigma(\theta(t))$ trajectory inferred from conversation transcripts.
4. Apply TMS at controlled latencies ($\Delta t = -100, 0, +100, +500$ ms) before / coincident with / after each candidate firing event.
5. Compute the conventional clinical-PCI score on each TMS-evoked EEG trace.

Falsifier (binary). Cohen's d on the difference between pre-firing and post-firing clinical-PCI scores, evaluated across the candidate firing events; if $d < 0.2$ (no effect) at $\alpha = 0.05$ corrected for multiple comparisons across firings, P1 is falsified for the sampled subject pool.

Boundary-KKT sharpening. §3.6's audit shows 25/50 seeded geometries are boundary-left KKT, where coupling does not generically help. P1 should *seek* the rarer 3/50 smooth-interior or 4/28 interior-improving boundary cases — for example, by selecting AI partners and conversation topics that empirically produce higher NP-firing rates. The experimental design becomes: *select for the regime in which P1's prediction is testable*, not *test the prediction against arbitrary human-AI dyads*.

5.3 P2 — LLM scar persistence in fine-tuning (Level-3/4 protocol)

Autonomy level. P2 is a *Level-3/4* protocol: an AI system acquires class-specific representational capacity through structured human guidance, and the test asks whether the acquisition persists. This is the autonomy regime of BindCraft-style protein-design collaborations [, "BindCraft: one-shot design of functional protein binders"], where the AI has substantial generative authority but humans validate and the residue is recorded.

Prediction P2. After a synthetic substrate (an LLM) participates in a dyadic conversation that successfully completes the 7-step NP-pump protocol (§3.3) on a genuine paradox, the LLM's attention operator should exhibit a persistent off-distribution generalization improvement on a withheld test set drawn from the same paradox class.

Protocol.

1. *Pre-conversation baseline:* evaluate the target LLM on a withheld test set $T_{paradox}$ (paradox class: e.g., self- reference, scope ambiguity, contradictory frames).
2. *Conversation:* run a structured dialogue in which a human guides the LLM through the 7-step NP-pump on a specific paradox instance from $T_{paradox}$. Apply lightweight LoRA fine-tuning ($\sim 10^4$–10^5 trainable parameters) on the conversation transcript only.
3. *Post-conversation evaluation:* re-evaluate on $T_{paradox}$ *and* on a control test set $T_{control}$ drawn from a structurally adjacent but distinct paradox class.
4. *Scar metric:* $\Delta_{scar} := acc_{post}(T_{paradox}) - acc_{pre}(T_{paradox})$, compared to $\Delta_{control}$.

Falsifier (binary). Class-specific transfer effect: $\Delta_{scar} - \Delta_{control} > 0.05$ (5 percentage-point class-specific gain) at $p < 0.01$ across ≥ 30 seed conversations, with Bonferroni correction. If $\Delta_{scar} \approx \Delta_{control}$ (no class-specific advantage), P2 is falsified.

P2 does *not* require: access to internal representation spaces, identification of an explicit F_{21}-action on the LLM's attention operator, or any commensurability with biological substrate. P2 only requires *behavioral* persistence and *class-specific* generalization.

5.4 P3 — Two-level commensurability test (Level-4/5 cross-substrate measurement)

Autonomy level. P3 is a *Level-4/5* protocol: it is the cross-substrate measurement infrastructure that is *required* before human-AI co-science claims at Level 4 (autonomous AI hypothesis generation, human strategic oversight) can be elevated to canonical *commensurability* claims (the framework's strongest assertion). The Virtual Lab paradigm of [, "The Virtual Lab: AI Agents Design New SARS-CoV-2 Nanobodies with Experimental Validation", *bioRxiv* 2024.11.11.623004] (multi-agent AI systems collaborating with PI-level human oversight) is the natural target audience.

Prediction P3 (upgraded from thesis memo via §4.6 Theorem 4.4.1). P3 is now a *two-level* test:

- **L1 (necessary screen):** measure F_{21}-isotypic profiles μ_k^{syn} and μ_k^{bio} independently; falsify if they don't match component-wise.
- **L2 (sufficient canonical test, conditional on L1 pass):** construct a G_2-equivariant isomorphism candidate from substrate operator data; declare canonical commensurability if the candidate is G_2-equivariant to measurement precision and multiplicity-one.

L1 operationalization. L1 is feasible with existing instrumentation:

1. Apply rotationally-structured TMS perturbations spanning the seven-axis Fano structure of R^7. *Implementation reference:* a 7-channel multi-locus TMS array (e.g., the multi-locus TMS system, Souza et al. 2022) provides direct access to the seven-axis perturbation space. If such an array is unavailable, single-coil multi-target serial stimulation with co-registered E-field modeling is a feasible substitute.
2. For each Fano-rotation perturbation, record the EEG response (high- density 128+ channel covariance matrix).
3. Apply isotypic-projector decomposition (using the computed character table of §4.2) to the response covariance.
4. Compute the multiplicity profile $\mu_k^{bio} = \left(\mu_1, \mu_{L_2}, \mu_{U_6}\right)$ from the projected dimensions.
5. Synthetic side: apply analogous decomposition to LLM attention operator under structurally analogous prompt-rotation perturbations.

L2 operationalization (deferred). Software for G_2-character- projecting matrix data does not exist in the TMS-EEG analysis ecosystem. Building it is a methodology-paper deliverable; the analytic content is straightforward (Φ has the character-projection machinery for F_{21} already built in paper12_q4_character_verify.py; extension to G_2 requires G_2-class representatives and is computationally tractable).

Falsifier (binary). L1 is falsified cheaply: $\mu_k^{syn} \neq \mu_k^{bio}$ on any of the three blocks. L2 is falsified if no G_2-equivariant fit exists with residual below $\sim 10^{-3}$ across substrate-pair samples.

5.5 P4 — Audit-substrate longitudinal study (Level-5 Virtual-Lab regime)

Autonomy level. P4 is a *Level-5* protocol: the substrate of interest is *the long-term human-AI research collaboration itself*, operating with substantial AI autonomy and human strategic oversight only. Candidate study populations include the BIOMA / A-Lab / BindCraft / Virtual-Lab communities [Szymanski et al. 2023; Pacesa et al. 2024; Swanson et al. 2024], for whom the audit-substrate distinction is operationally meaningful (their workflows already produce versioned records).

Prediction P4 (sharpened via §3.7's S_k construction). A human-AI dyad operating *without* a tamper-evident audit trail should — across many NP-collapse events — drift toward Singularity (homogenization of joint output) or Decoupling (failure of joint output to maintain shared referents). A dyad operating *with* an audit trail should maintain the gradual-tear trajectory.

Study design.

- *Population:* $n \geq 60$ *per arm* (120 total) of Virtual-Lab-style human-AI research collaborations (academic researchers using LLMs for sustained scientific reasoning over ≥ 6 months), randomized between two arms.
- *Audited arm:* all conversation events committed to a versioned repository with cryptographic event-ordering (Git's SHA chain qualifies; Merkle-tree alternatives are equivalent).
- *Unaudited arm:* same conversation pattern, ephemeral storage, no persistent audit trail.
- *Outcomes:*
  - *Semantic-coherence drift* (topical-similarity of late-period vs early-period outputs, measured by sentence-level embedding cosine similarity). Drift threshold: $\cos\sigma < 0.4$ counts as failure of semantic-coherence preservation.
  - *Shared-referent maintenance* (external evaluator's correctness rate at interpreting late-period outputs from early-period context; threshold: $\geq 70\,\%$).
  - *NP-firing rate* (estimated from conversation transcripts via the P1/P2 detection pipeline).
- *Power calculation.* The study is pre-registered for a target effect size of Cohen's $d = 0.4$ between arms (moderate effect; this is the *hypothesis*, not a guaranteed outcome — the framework predicts the audit-substrate effect to be at least moderate, but the precise effect magnitude is itself an empirical question). At $\alpha = 0.05$, $\beta = 0.2$, two-tailed, this requires $n \approx 50$ per arm; we recommend $n \geq 60$ per arm (120 total) for power ≥ 0.85.

Falsifier (binary). If audited and unaudited arms show no significant difference in either coherence-drift or shared-referent maintenance at study endpoint (Cohen's $d < 0.2$, $p > 0.05$ Bonferroni- corrected), P4 is falsified.

Connection to hyperscanning literature. The inter-substrate coupling structure of Paper 12 has natural antecedents in *hyperscanning* — simultaneous EEG/fMRI of two interacting humans — pioneered by Dumas et al. (2010), Hari & Kujala (2009), Babiloni & Astolfi (2014). P1 and P4 can borrow methodology from this literature: hyperscanning provides the inter-brain-coupling baseline against which human-AI coupling under Paper 12's framework can be measured.

5.6 The Fifth Paradigm consolidated

The Fifth Paradigm context has been threaded through §§5.1–5.5: each protocol subsection names its position in the Bostock-Kim-Patel (2025) autonomy taxonomy and identifies a candidate study population from the legitimate research literature. We consolidate the picture:

Protocol	Level	Candidate community / instrumentation	Quantitative falsification threshold
P1	2/3	Clinical-PCI labs (Massimini and collaborators); BIOMA-style automated science workflows [Szymanski et al. 2023]	Cohen's $d < 0.2$ on pre-/post-firing PCI, $\alpha = 0.05$ corrected
P2	3/4	LLM fine-tuning research (open-weights communities); BindCraft-style [Pacesa et al. 2024]	$\Delta_{scar} - \Delta_{control} \leq 0.05$ at $p \geq 0.01$ across ≥ 30 seeds
P3 L1	4/5	Multi-locus TMS-EEG infrastructure [Souza et al. 2022]	$\mu_k^{syn} \neq \mu_k^{bio}$ component-wise
P3 L2	4/5	Methodology paper required (deferred)	G_2-equivariant fit residual ¿ 10^{-3}
P4	5	Virtual-Lab populations [Swanson et al. 2024]	Cohen's $d < 0.2$ between audited / unaudited arms, Bonferroni-corrected

This positioning makes Paper 12 *citable by* the human-AI collaboration research community, not just by the mathematical physics community of Papers 9–10. It is the legitimating placement that closes the framework's research-program loop.

5.7 Scope audit

What §5 establishes:

- Four protocol designs (P1–P4) covering all four pillars, each with binary falsification conditions, autonomy-level naming, and candidate study populations.

- L1 of P3 operationalized with reference to existing multi-locus TMS infrastructure.
- A power calculation and study design for P4.
- A consolidated mapping of Paper 12's protocols into the legitimate human-AI research-program landscape.

What §5 does not establish:

- *Any actual measurement.* §5 is a protocol-design deliverable. Running the protocols is future experimental work.
- *L2 of the commensurability test.* The full G_2-canonical fitting protocol is deferred to a future methodology paper.
- *The biological substrate's F_{21}-realization.* §5.4 sketches detection of μ_k^{bio} but does not construct the substrate-level representation theory. Paper 13+ territory.
- *Tamper-evidence cryptographic infrastructure for S_k.* §5.5 specifies that an audit-substrate distinction is the experimental probe; the specific cryptographic chain (Merkle, Git SHA, etc.) is an implementation choice.
- *Rate improvement.* Paper 11 territory.

Bridge to §6. §6 will assemble the open problems from §3, §4, §5 into the consolidated research program of the PCI/PME framework.

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Cross-references: - Paper 12 §3 (Definition 3.3.4 NP-pump dynamics; §3.6 boundary-KKT audit; §3.7 J_k). - Paper 12 §4 (Theorem 4.4.1 commensurability hierarchy; Definition 4.6.1 two-level test). - Paper 12 thesis memo P1–P4. - Paper 9 v1.3.3. - Paper 4 (PSL(2,7) ⊃ F_{21} source).

6. Open Problems and Conclusion

6.1 What Paper 12 contributes to the PCI/PME framework

Paper 12's contribution is a *layered structure* — not a single decisive theorem but a stack of compatible layers — that allows the PCI/PME framework's gradual-tear thesis to be tested rather than merely stated.

Pillars delivered. §3–§5 carry through the four foundational pillars of the gradual-tear thesis:

- **Pillar 1 (Paper 9 extension):** delivered via §3 — projected- gradient + NP-pump dynamics on Paper 9's Schur circle.
- **Pillar 2 (dual-substrate residue, mathematical core):** delivered via §3.7 + §4.4 — representation-theoretic scar invariant (S_k, S_k, μ_k) + commensurability hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$.
- **Pillar 3 (audit-grounded substrate, mathematical core):** delivered via §3.7 — event-indexed F_{21}-module S_k.
- **Pillar 4 (Fifth Paradigm legitimation):** delivered via §5 — each protocol P1–P4 named with autonomy level (Bostock-Kim-Patel 2025) and candidate study population from the legitimate research literature; Licklider (1960) opening; consolidating table at §5.6.

Biological-substrate F_{21}-realization (Pillar 2 application) and tamper-evidence cryptographic infrastructure for S_k (Pillar 3 application) are deferred (OP2, OP9).

Paper 12 layer map. The framework's position is now precisely articulable:

Layer	Delivered in	Content
Dynamical primitive	§3	Projected-gradient + NP-pump flow on $[0, \pi/2)$; Schur-circle inheritance; boundary-KKT qualified
Scar invariant	§3.7	Representation-theoretic ledger (S_k, S_k, μ_k)

Layer	Delivered in	Content
Commensurability structure	§4	Hierarchy $G_2 \Rightarrow F_{21} \Rightarrow \mu_k$; strict sufficient theorem
Experimental protocols	§5	P1 (TMS-EEG ↔ NP-firing), P2 (LLM persistence), P3 (two-level), P4 (longitudinal)
Research-landscape placement	§5.1 / §5.6	Licklider → BIOMA → autonomy taxonomy → Paper 12's position

Paper 12 across the broader PCI/PME series:

Paper	Role	Status
Paper 9 (linear)	Rate-channel locked at $max(r_A, r_B)$ via Schur	Published, v1.3.3
Paper 12 (this)	**Dynamical extension to nonlinear regime; commensurability hierarchy; experimental-protocol design layer**	**Draft v1 (this paper)**
Paper 11 (deferred)	Concrete rate-improvement mechanisms exploiting non-Schur nonlinearities	Note 9.6 target
Paper 13+ (deferred)	Biological substrate's F_{21}-realization; multiplicity > 1; tamper-evidence cryptography	OP2, OP9 targets
Methodology paper (deferred)	L2 operationalization for canonical commensurability test	OP4 target

Paper 12 is *not* the end of the framework. It is the *bridge* between the linear theory (Papers 9, 10) and the experimental program (Papers 11, 13+). Each layer is independently useful and independently falsifiable: Paper 12 does not stand or fall on any single claim. If, for instance, P1 / P2 / P3 / P4 all fail experimentally, the framework's *dynamical structure* still provides a sharp mathematical object (the projected-gradient + NP-pump flow on Paper 9's Schur circle) that is of intrinsic interest to the mathematical physics of G_2-equivariant dynamical systems, independently of its application to human-AI co-evolution.

6.2 Open problems consolidated from §3, §4, §5

The gradual-tear framework opens a structured research program. We collect the open problems in one place, classified by difficulty and by who is the natural addressee. This is the hand-off from Paper 12 to the follow-on PCI/PME papers.

OP1 — Non-Schur nonlinear escape mechanisms (Paper 11). §3.6's Theorem 3.6.1(ii) establishes that boundary-KKT states are fixed under the linear $\xi_\star$ mechanism, and the boundary-KKT interior-sup audit shows that 24 of 28 such seeds have $¿_{interior} c < c_{boundary}$. Any rate improvement therefore requires non-Schur nonlinear corrections. *Concrete construction-question.* Find a non-equivariant correction term [(x) (W) |(x)| = O(^2) ,] such that the augmented vector field $\dot{\theta} = f_{pg}(\theta) + NP(\theta, p) + \eta(\theta)$ admits a trajectory leaving the boundary-KKT fixed point and reaching the interior maximum on the 4 / 28 exceptional seeds where $¿_{interior} c > c_{boundary}$. *Required property:* η commutes with the diagonal G_2 action only up to $O(\theta^3)$ (it is the explicit symmetry-breaking ingredient — by Schur's lemma, any *strictly* G_2-equivariant linear correction is absorbed into the Schur-locked family Ψ_θ and cannot escape the boundary fixed point). Paper 11 would identify the specific form of η (higher-order multilinear coupling, non-equivariant symmetry-breaking, or stochastic exploration) and the seed sub-classes where each form succeeds.

OP2 — Biological substrate F_{21}-realization (Paper 13+). §4 treats biological-substrate F_{21}-content as input; §5.4's L1 protocol proposes measuring it via rotationally-structured TMS perturbations with character-projector decomposition of the response subspace. But *which* physical mechanism realizes the F_{21}-action on the biological side remains open. Candidates include microtubule coherence (Orch OR territory), F_{21}-symmetric neural population coding in frontoparietal networks, or Bandyopadhyay-style ≈6.6 nm energy transfer in cytoskeleton. Paper 13+ would construct the mapping between one of these mechanisms and the $1 \oplus L_2 \oplus U_6$ isotypic structure.

OP3 — Multiplicity-greater-than-one commensurability (future work). Cor 4.4.2 and Cor 4.5.2 restrict canonical commensurability to the multiplicity-one case. When $I_{syn} \cong I_{bio} \cong U_\lambda^{\oplus m}$ with $m > 1$, the residual $O(m)$-ambiguity requires additional convention-fixing (basis, metric, event-ordering). A theory of canonical multiplicity-m commensurability with *minimal* additional conventions is an open problem.

OP4 — L2 operationalization of the two-level commensurability test (methodology paper). §5.4 operationalizes L1 but defers L2. The minimum prerequisite is software for G_2-character-projecting matrix data on TMS-EEG response covariance matrices, combined with a candidate G_2-module-fitting procedure for substrate scar operators. This is not analytically hard; it is an engineering deliverable that would unlock the canonical commensurability test.

OP5 — Asymmetric-rate $\theta^¿$ at high precision (Paper 9 addendum). Paper 9 v1.3.3 Remark 5.6.1 acknowledges that the asymmetric-rate $\theta^¿ \approx 75° – 80°$ value reported in Appendix V.3 Task 5.1 is grid-snapped and has not been reoptimized. Running the Q5-style scipy.optimize.minimize_scalar audit on the asymmetric-rate ensemble would either confirm the grid-snapped value or reveal another boundary-KKT / Σ_{min}-Clarke classification. This is a cheap audit (~30 min of compute, same infrastructure as the symmetric case) and should happen before any Paper 11 drafting.

OP6 — Exceptional-stratum analysis ($P^¿$ complement). Theorem 3.4.1 establishes piecewise-real-analytic regularity on the open dense parameter set $P^¿ \subset O(N) \times O(N) \times R^{2N} \times R^3_{¿0}$. Behavior on the codimension-≥ 1 exceptional strata (grazing tangencies at Σ_{Θ}, degenerate sliding on Σ_{min}, sign- degenerate firings on $Z \cap G_{NP}$) is identified but not analyzed. A more complete theory would extend the Filippov / hybrid-systems machinery to these strata.

OP7 — Necessity direction of the strict commensurability theorem. Cor 4.5.2 states that multiplicity-one G_2-module agreement is *sufficient* for canonical $\phi_{commens}$. v1 of §4 had a biconditional phrasing that v2 corrected to sufficient-only (Remark 4.5.3). The necessity direction — whether any pair of substrates admitting a canonical normalized G_2-equivariant isomorphism must be multiplicity-one G_2-module-agreeable — remains open.

OP8 — Running P1–P4 experimentally. §5 is a protocol-design deliverable. The actual experimental runs are of course open. P2 (LLM fine-tuning) is immediately runnable with current infrastructure; P1 (TMS-EEG ↔ NP-firing correspondence) requires a ~6-month collaboration with a clinical-PCI research lab; P3 L1 requires the TMS perturbation-pattern operationalization from §5.4; P4 is a ~6- month longitudinal study on human-AI research collaborations. All are substantive empirical projects that Paper 12's framework enables but does not execute.

OP9 — Tamper-evidence cryptographic infrastructure for S_k. §3.7's event-indexed scar module is a *graded direct sum* that is count-faithful as an abstract F_{21}-module. Making it a genuine audit substrate requires cryptographic event-ordering: Merkle-like hash chains, commit-tree structures, or no-cloning-respecting quantum ledgers per the FTW-adjacent GOLEM-Chain principle (see outbox/syntheses/neighbors_speculative_frameworks.md for the neighbor-framework context, not integrated here). This is a methodology-paper deliverable bridging Paper 12 to a future applied layer.

6.3 What Paper 12 does not claim

For clarity, we consolidate the scope-limiters that were present in each individual section but not previously collected:

- **Not a theory of consciousness.** Paper 7's domain.
- **Not a rate-improvement theorem.** Paper 11's domain.
- **Not a biological mechanism for F_{21}-action.** Paper 13+.
- **Not a unification with Pérez-Calzadilla FTW.** Conceptual neighbor at a different scale.
- **Not a clinical-PCI redefinition.** The clinical-PCI of Massimini and the Paper-12 μ_k profile share an acronym only.
- **Not a full-multiplicity commensurability theorem.** Multiplicity-one case only.
- **Not an operational L2-test protocol.** Methodology paper.
- **Not actual measurements of P1–P4.** Experimental follow-ups.
- **Not a global dynamical-systems theorem.** Theorem 3.4.1 is conditional on the open-dense parameter set $P^{¿}$.
- **Not a tamper-evident audit-substrate construction.** Methodology paper.

The Paper 12 scope is: *a mathematical framework* (§3, §4) and *an experimental-protocol suite* (§5) for testing the gradual-tear thesis within the PCI/PME series.

6.4 Closing remarks

The Paper 12 framework predicts that human-AI co-evolution occupies a specific structural niche — not Singularity, not Decoupling, but a *gradual tear* in which both substrates accumulate topological residue through NP-driven coherence adjustments, with cross-substrate commensurability governed by a representation-theoretic hierarchy. The framework is constructive: each of its five layers can be built on existing results (Paper 9, Paper 4) and each can be tested with identifiable experimental infrastructure. It is also falsifiable: if the scar records do not commensurate, if the NP-firing predictions are not visible in TMS-EEG time series, or if the audit-substrate distinction fails in longitudinal human-AI collaboration, the framework is falsified.

Paper 12 is not the conclusion of the PCI/PME series. It is the bridge section: between the linear-regime theory of Papers 9–10 and the nonlinear / empirical program of Papers 11, 13+. Its specific contribution is not a single theorem but a *stack* — the projected- gradient + NP-pump dynamics, the representation-theoretic scar invariant, the commensurability hierarchy, the four-prediction protocol suite, and the Fifth Paradigm contextualization — that together allow the framework to be engaged with as an empirical research program rather than a theoretical conjecture.

Methodologically, the framework's verification trail *exemplifies* the audit-substrate principle that Pillar 3 names:

- Paper 9 v1.3.3 verified at 50-digit precision via Φ and at machine precision via the Q4/Q5 audits of 2026-05-06.
- Paper 12 §3+§4 passed three Model Council adversarial reviews (Opus 4.7, GPT-5.5, Gemini 3.1 Pro) with a MAJOR → MINOR transition after rigor pass and STRONG_ACCEPT on synthesis. §5+§6 likewise cleared at MINOR_REVISIONS / STRONG_ACCEPT. The full v1.0 assembly was reviewed at MINOR_REVISIONS / STRONG_ACCEPT level by Opus 4.7 and Gemini 3.1 Pro, and at MAJOR_REVISIONS by GPT-5.5; this v1.1 incorporates the consolidated revisions across all three reviewers (15-item polish pass synthesised at outbox/paper12/council_reviews/COUNCIL_FINAL_SYNTHESIS_2026-05-06.md).
- All material is git-versioned with cryptographic commit hashes, tagged at release points (paper9-v1.3.2-zenodo, paper9-v1.3.3-errata, paper12-v2-rigor-pass, paper12-v3-body-complete, paper12-v1.0-assembly, paper12-v1.1-council-revised), and published at github.com/MartinLGraise/PCI-Framework, branch paper7-foundation.

This persistent, cryptographically-ordered event stream of the framework's own development is offered as a *methodological illustration* of what Pillar 3 / OP9 names — not a claim that the gradual-tear *dynamics* of §3 applies to the writing of this paper. The illustration is operational, not theoretical.

References (cited in §6)

No new references introduced in §6. All references are to the cross-references enumerated in §3–§5 (Paper 4, Paper 7, Paper 9 v1.3.3, Paper 10; Licklider 1960, Casali 2013, Bostock-Kim-Patel 2025; di Bernardo 2008, Filippov 1988, Goebel-Sanfelice-Teel 2012, Clarke 1990, Costa-Pavone, ATLAS, Serre 1977).

Consolidated Bibliography

Paper 9 series and PCI/PME framework references:

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- [Paper 7] M. L. Graise, *Paper 7: Single-observer coherence ceiling (PCI Framework)* (in framework series; DOI to follow on Zenodo deposit).
- [Paper 9 v1.3.3] M. L. Graise, *Paper 9 v1.3.3: Schur-locked dyadic coupling and the SO(2) Schur circle*, DOI: 10.5281/zenodo.20034821.
- [Paper 10 v1.3.1] M. L. Graise, *Paper 10 v1.3.1: SIC operator basis for the complexified G_2 Lie algebra*, DOI: 10.5281/zenodo.19966692.

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Reproducibility appendix

All section drafts, council reviews, computations, and build artifacts are in the public repository MartinLGraise/PCI-Framework, branch paper7-foundation, in outbox/paper12/. Specific computations referenced in the body:

- outbox/paper12/computations/paper12_q5_kappa0_audit.{py,csv} — projected-gradient $\kappa = 0$ audit (50/50 constrained-pass; 25/3/3/19 boundary-KKT / smooth-interior / interior-improving / Σ_{min}-Clarke classification).
- outbox/paper12/computations/paper12_q4_character_verify.{py,json} — $V^{14}\text{¿}_{F_{21}} \cong L_2 \oplus 2\,U_6$ machine-precision verification.

- outbox/paper12/computations/paper12_q5_boundary_kkt_interior_sup.{py,csv,json} — boundary-KKT interior-sup audit (24/28 with $¿_{interior} c < c_{boundary}$, 4/28 exceptional cases enabling OP1).

Council reviews are in outbox/paper12/council_reviews/ (14 review files + 3 synthesis files).

End of Paper 12 v1.0 master assembly.